

Part V — Used serviceable material and teardown

Descriptive reference. Sources are cited inline and listed at the end. Takes no view.

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PART V

Used serviceable material and teardown

This Part explains where used engine parts come from, what has to be true before one may be installed again, what a retired aircraft or engine is worth when it is taken apart, why the supply of such parts depends on retirements that did not happen, how a used part is priced against a new one, who the traders are, and why the module is the unit in which this material increasingly trades. A reader who finishes it will be able to read a teardown quote, a used-part listing and a used-material market estimate with every term understood.

TRANSLATION-TABLE ROWS ANSWERED

This Part answers the rows of the Part III translation table that run through the supply of used serviceable material (USM), meaning parts made by the original equipment manufacturer (OEM), taken from retired aircraft and engines, and released for reinstallation. (Part III §24, row T-1: low retirements): fewer retirements mean fewer aircraft and engines torn down, so the supply of used parts and of part-worn modules is smaller than the shop-visit demand of Part IV implies; §33 carries the feedstock arithmetic. (Part III §24, row T-6: lease extensions): an aircraft whose lease is extended is not returned, not parted out and not a source of material; §32 and §33. Two rows name the new-generation engines, the geared turbofan (GTF) from Pratt & Whitney and LEAP (CFM's new-generation engine), and one names the 737 MAX (Boeing's successor to the 737NG). (Part III §24, row T-2: GTF groundings) and (Part III §24, row T-3: LEAP durability): the old fleet kept flying consumes used parts at shop visits while releasing none; §33. (Part III §24, row T-4: 737 MAX production cap) and (Part III §24, row T-5: Airbus output): without a replacement aircraft the retirement decision of §33 is not taken, whatever the part-out value; §32 and §33. (Part III §24, row T-7: OEM parts escalation): used parts are priced as a share of the catalogue list price of the original equipment manufacturer (OEM), so every list increase lifts the used price too; §34. (Part III §24, row T-11: freighter conversions): a conversion candidate needs engines with life, so conversion competes with teardown for the same airframes and engines; §32.

31. What makes a used part serviceable

A used part is not a lesser kind of part. It is the manufacturer's own part, taken from an engine or aircraft that has been in service, and put through a process that ends with a signed release. The process is what this section describes. At the end of it the part is "serviceable", a word with a legal meaning, and it can be bolted into another engine. Without that process the same piece of metal is scrap.

USED SERVICEABLE MATERIAL (USM)

Used serviceable material is any aircraft or engine part that has been in service, has been removed, and has been inspected (and repaired if needed) to the manufacturer's manual and released on an airworthiness tag with its full history, so that it is legally fit to be installed again. It is OEM-made, so it sits outside the debate over alternative-manufacturer parts (Part VI), but it competes for the same line in a shop visit's parts bill. A part with a gap in its paperwork is scrap, whatever its physical condition. Its supply is a lagged function of retirements (§33). [AJW Group (AJW), an engine-parts company, via AviTrader, Aug and Oct 2025; D3 §2.1, the USM dossier behind this Part (D3)]

The alternatives to a used part, when a shop needs to replace one, are three. A new OEM part is bought from the manufacturer at its published catalogue list price (CLP), the price Part I §4 glossed and Part IV §28 owns. An alternative-manufacturer part is one made by a third party under a Parts Manufacturer Approval (PMA) from the US Federal Aviation Administration (FAA); Part VI owns it. A repair is the third alternative, either one the manufacturer's manual describes or one approved separately by an FAA Designated Engineering Representative (DER), which Part VI §40 owns. This Part is about the first substitute only, the used part. [D3 §2.1; D4 §2.1, the alternative-parts dossier (D4)]

31.1 HOW A PART LEAVES SERVICE, AND THE THREE PLACES IT CAN GO

A part leaves service in one of three ways. It is removed at a shop visit, when the engine is open and inspection finds it out of limits or finds a better part to install. It is removed from an aircraft at retirement. Or the whole aircraft or engine is torn down, which the trade calls part-out. Part I §8 boxed part-out and teardown: part-out is the owner's decision to dismantle and sell the components; teardown is the physical disassembly. §32 carries the economics. [D3 §2.1; Part I §8]

From there each part goes one of three ways, and the three outcomes are the vocabulary of every teardown report.

SERVICEABLE AS REMOVED

A removed part is serviceable as removed when it has been inspected to the applicable manual and found within limits. It is tagged and sold without repair. It is the cheapest outcome for the trader, because there is no repair bill, and the fastest, because there is no repair-shop queue. [D3 §2.1]

The second outcome is the repairable part. It is outside the manual's limits but can be brought back inside them. For an airframe component the limits and the permitted repairs are in the Component Maintenance Manual (CMM), the manufacturer's document for that component, which Part I §9 boxed. For an engine part they are in the Engine Shop Manual (ESM), also boxed in Part I §9. A repair the manual does not cover may be done under a separately approved DER scheme (Part VI §40). The part goes to a repair shop, is repaired, tested, tagged and sold. The repair cost is the main cost of goods for a trader in used material. [D3 §2.1]

BEYOND ECONOMIC REPAIR (BER)

A removed part is beyond economic repair when it is outside limits and not worth repairing: the repair would cost more than the part would fetch, or no approved repair exists. It is scrapped. [D3 §2.1]

LIFE-EXPIRED

A life-limited part is life-expired when it has reached the cycle limit the manufacturer certified for it. It has no used-material value, however good its condition, and it must be mutilated so that it cannot re-enter the supply chain. Part I §4 carries the limits: 20,000 cycles for the -7B's compressor and turbine disks, 25,000 for the low-pressure turbine, 30,000 for the fan and booster. [D3 §2.1; Part I §4]

EXHIBIT 5.1 — THE THREE OUTCOMES FOR A REMOVED PART, AND WHAT EACH MEANS FOR THE TRADER

Outcome	Test	What happens next	Cost to the trader	Time to sale
Serviceable as removed	Inspected to the CMM or ESM; within limits	Tagged by an approved organisation; sold	Inspection and tagging only	Immediate once tagged
Repairable	Outside limits; a repair exists in the CMM or ESM, or under an approved DER scheme	Sent to a repair shop; repaired; tested; tagged; sold	The repair bill, the main cost of goods in used material	After repair-shop turnaround
Beyond economic repair, or life-expired	Outside limits and not worth repairing; or a life-limited part at its cycle limit	Scrapped; a life-expired life-limited part is mutilated	Disposal	None

Source: AJW Group / AviTrader, "Ensuring the airworthiness of USM", Aug 2025 (PDF) and AviTrader, 16 Oct 2025; D3 §2.1. "DER" is a Designated Engineering Representative (Part VI §40).

The exhibit explains a fact that recurs throughout this Part. Most used material is not serviceable as removed. It passes through a repair shop and is re-certified before it can be sold, so the capacity and price of repair shops (Part IV §27) set both the cost and the pace of the used-parts trade. Rising repair costs are named alongside falling supply as a driver of CFM56 (the engine of Part I) and V2500 (the alternative A320ceo engine of Part I §10) part prices (§34). [Aviation Week, "Magnetic expects CFM56 market challenges", 2024; D3 §5]

31.2 THE RELEASE TAG

A part becomes serviceable in the legal sense when an approved organisation signs it off. Part I §9 boxed the two forms and the release tag itself, so this is a reminder. In the FAA system the form is FAA Form 8130-3, the Authorized Release Certificate, also called the Airworthiness Approval Tag. For a repaired or inspected used part it is issued by a repair station certificated under Part 145 of the US regulations; for a new part it is issued by the holder of a production approval. In the European system the equivalent is Form 1 of the European Union Aviation Safety Agency (EASA), issued by an EASA Part-145 maintenance organisation or a Part-21

production organisation. Both certify that the part complies with the applicable regulations and is safe for use. [Rotabull, 8130-3 instructions, undated; AJW Group via AviTrader, 2025]

Many repair stations hold both approvals and issue a dual release, one tag valid in both systems. That matters commercially. A part tagged for one system only cannot be installed on an aircraft registered in the other without re-certification, so cross-border trade in used parts runs through dual-release shops. [D3 §2.1, §5]

The tag is necessary and not sufficient. Two further things decide whether a buyer will accept the part: its documented history (§31.3) and the identity of the organisation that signed the tag. The trade press is blunt about the second: the 8130-3 and Form 1 documents "can be easily generated", so buyers vet the supplier as well as the paperwork. Delta's published supplier matrix for aftermarket parts is one example of an airline codifying which supplier categories and which tag types it accepts. [AJW Group via AviTrader, 2025; Delta TechOps supplier matrix, Mar 2024]

31.3 TRACEABILITY: WHY A PAPERWORK GAP SCRAPS A PART

No regulator allows "untraceable" used parts. Everything must be documented, certified and compliant. [AJW Group via AviTrader, 2025] The documentation package a buyer requires differs by part class, and the class that matters most for engines is the one with the strictest rule.

HARD-TIME (HT) PART

A hard-time part is one with a fixed overhaul or replacement interval, as distinct from an on-condition part that is replaced when inspection finds it worn. It needs complete operating-history records and evidence of compliance with every applicable Airworthiness Directive and Service Bulletin. [AJW Group via AviTrader, 2025; D3 §2.1]

Part I §3 boxed the two instructions just named. An Airworthiness Directive (AD) is a regulator's order making a fix or inspection mandatory; a Service Bulletin (SB) is the manufacturer's instruction to modify or inspect. Compliance records for both must travel with hard-time parts and with life-limited parts, and an AD inspection can itself be the clock that ends an engine's remaining usable time (Part II §16). [Part I §3; D3 §2.1]

For life-limited parts the standard is the one Part I §9 boxed as back-to-birth (BTB) traceability: an unbroken record of every cycle the part has accumulated since new, on which engine serial numbers and under which operators. The arithmetic is why the rule is absolute. Remaining life is the manufacturer's limit minus the documented cycles consumed. An undocumented period cannot be assumed to contain zero cycles. So a gap means the part cannot be shown to have any life, and a part that cannot be shown to have life has no value. [D3 §2.1; Part I §9]

WHY A PAPERWORK GAP SCRAPS A PART

Consider a high-pressure turbine disk with a 20,000-cycle limit whose records show 11,000 cycles on three engines, then an eighteen-month period with no record, then 2,000 more cycles. The physical disk may have flown 13,000 cycles and have 7,000 left. But nobody can prove the eighteen months contained no flying. A buyer's engineer cannot write a remaining life into the receiving engine's records, so the part cannot be installed; the shop manual's remaining-life calculation has no input. The part is physically present and commercially absent, exactly as Part I §3 said of a module with a gap in its records. The usable supply of used life-limited parts is the documented supply, not the physical one. [D3 §2.1; Part I §3, §9]

Other documents are standard in a used-part package whatever the class: the removal tag from the last operator, the teardown report, a certificate of conformance from the seller, and a declaration that the part has never been through an accident.

NON-INCIDENT STATEMENT

A non-incident statement is the seller's declaration that a part was not installed on an aircraft involved in an accident, a fire or an immersion. It is a standard element of a used-material documentation package, beside the removal tag, the teardown report and the certificate of conformance. [D3 §2.1]

What makes a part unsaleable follows from the above. Missing documents mean the part will not be used. An unclear maintenance history or gaps in the paperwork are grounds for rejection. Mismatched part or serial numbers, unusual formatting on a tag, or an unfamiliar issuing organisation all trigger quarantine. [AJW Group via AviTrader, 2025]

QUARANTINE

Quarantine is the segregation of a part whose documentation is suspect, pending investigation. The triggers named in the trade press are mismatched part or serial numbers, unusual formatting on a release tag, and an issuing organisation the buyer does not recognise. [AJW Group via AviTrader, 2025; D3 §2.1]

EXHIBIT 5.2 — THE USED-PART DOCUMENTATION PACKAGE, BY PART CLASS: WHAT IS REQUIRED, WHO ISSUES IT, AND WHAT A GAP DOES

Part class	Documents required	Who issues them	Effect of a gap
Any used part	Release tag (FAA 8130-3 or EASA Form 1, dual release for both markets); removal tag from the last operator; teardown report; certificate of conformance; non-incident statement	Approved repair station or production approval holder; last operator; teardown company; seller	Rejection; quarantine if the tag's numbers, format or issuer look wrong
Hard-time part	All of the above plus complete operating-history records and AD and SB compliance records	Every operator and shop in the part's history	Rejection: the next interval cannot be set
Life-limited part	All of the above plus back-to-birth traceability: every cycle since new, by engine serial number and operator	Every operator and shop in the part's history	The part has no provable remaining life and no value; at its limit it is mutilated
Module (Part I §3; §36 below)	The above for every life-limited part inside it, plus the module's own serial number, build standard and repair records with their approval basis	Shops and operators in the module's history	The module cannot be installed; its parts may still be sold individually if their own records are complete

Source: AJW Group / AviTrader, Aug and Oct 2025; Rotabull, 8130-3 instructions, undated; Delta TechOps supplier matrix F-840-018-A, Mar 2024; D3 §2.1; Part I Exhibit 1.12. "AD" and "SB" are Airworthiness Directive and Service Bulletin.

31.4 LIFE-LIMITED-PART RECORDS IN PRACTICE

Everything in §31.3 converges on the life-limited part, because it is the part whose used value is highest and whose paperwork rule is strictest. A used disk with life is the direct substitute for a new one at list price, and the next shop visit must replace any disk without enough life for the following run (Part I §4.2). So the trade advertises used life-limited parts on their documented remaining cycles, as Part I §4 showed with listings such as a core package advertised with 6,235 cycles remaining (CR). The records summary that supports such a listing is the mini-pack Part I §4.1 boxed: each life-limited part with its certified limit and its cycles consumed. [Part I §4; D3 §2.2]

Two consequences for a teardown follow. First, the documented fraction of an engine's life-limited parts is a hard cap on what the teardown can yield; §32 returns to this. Second, when a module changes hands as a unit, its documentation must show the history of every life-limited part inside it, which is why §36 can treat the module as an engine in miniature. [D3 §2.7, §5]

32. Teardown and part-out economics

Part-out is a decision; teardown is the work that follows. This section explains how a retired aircraft or engine is bought, where its value sits, how long it takes to turn into cash, and what the one set of published engine figures says. It ends with the full worked example the outline asks for: a 737-800 with two CFM56-7B engines, with every input labelled sourced or unknown.

32.1 HOW A RETIRED ASSET IS BOUGHT: PURCHASE AND CONSIGNMENT

Part I §3 boxed feedstock as the trade's word for the aircraft and engines bought, or taken on consignment, to be torn down for parts. Those are the two commercial routes, and they allocate capital and risk differently. [D3 §2.2; Part I §3]

Under purchase the teardown company buys the asset outright and owns the parts. It bears the capital cost and the inventory risk and keeps the margin. Independent traders such as AerSale and GA Telesis (GA), and the trading arm of AAR (AAR Corp., a parts distributor and maintenance company), do this. Lessors such as Willis Lease and AerCap feed their own retired lease assets into their own material subsidiaries, which is purchase by another name. [D3 §2.2]

CONSIGNMENT

Consignment is teardown and sale of parts on the owner's behalf for a fee or a commission, with title to the parts retained by the owner, who may be an airline, a lessor, a bank or an insurer. It is the alternative to buying feedstock outright and bearing the inventory risk. Willis Aero describes its business as "the acquisition or consignment of aircraft and engines"; the APOC Aviation (APOC) teardown of a 737-800 with Willis Lease offered "packages of parts available for sale and consignment". [Willis Lease Finance, 10-K FY2025; Asian Aviation, 2025; D3 §2.2]

32.2 WHERE THE VALUE SITS

The engines dominate the part-out value of a mature narrowbody. The engine figures in §32.4 put two run-out engines' used-material yield at \$3.2–4.4M; the single largest airframe item, the landing gear, fetches "over \$1 million" as a fully overhauled set, according to the magazine of ISTAT (the International Society of Transport Aircraft Trading). [Safe Fly Aviation, consultancy blog, 2026; ISTAT Jetrader, Autumn 2020] A sourced percentage split between engines and airframe was not captured by the research and is recorded as unknown U-D3-01. [D3, unknown U-D3-01]

Within the airframe the value sits in the rotables.

ROTABLE

A rotatable is a part that is repaired and reused rather than consumed: landing gear, the auxiliary power unit, thrust reversers, avionics boxes, flight-control actuators, wheels and brakes. Rotables are the airframe items a teardown sells; consumables and expendables are not worth the paperwork. [D3 §2.2]

AUXILIARY POWER UNIT (APU)

The auxiliary power unit is the small gas turbine in the aircraft's tail that supplies electrical power and compressed air on the ground and in some flight phases. It is one of the highest-value airframe rotatables in a part-out and one of the components for which lessors collect maintenance reserves (Part II §13). Its value in a 737-800 part-out was not captured by the research (U-D3-10). [D3 §2.2, unknown U-D3-10; Part II §13]

Within an engine the value sits in five places. First, life-limited parts with remaining cycles: fan, booster and compressor disks, shafts and spools, because the next shop visit must replace any that lack life for the following interval and a used disk with life substitutes directly for a new one at list price. Second, the high-pressure turbine (HPT) blades and nozzles and the high-pressure compressor (HPC) airfoils, the most expensive repairable parts; Part I §3 and §7 set out why. Third, fan blades. Fourth, cases and structural parts. Fifth, accessories, of which the fuel-metering unit is the example the trade names.

HYDRO-MECHANICAL UNIT (HMU)

The hydro-mechanical unit is the engine's fuel-metering unit, the accessory that sets fuel flow under the engine control's command. It is a high-value engine accessory and, with CFM56 fan disks and structural engine parts, one of the most searched-for teardown items in one 2026 consultancy report. [Safe Fly Aviation, teardown report, 2026 (consultancy blog); D3 §2.2]

32.3 TIME TO MONETISE, AND TEARDOWN AS THE END OF A LEASE

A 737-800 teardown takes about three months. During that time every part is assessed by technical teams before being sent to specialist repair shops for overhaul, testing and re-certification. Landing gear are among the first serviceable parts to become available. The auxiliary power unit is sold, and parts are offered as packages for sale and consignment. [Asian Aviation and AirlinerGS on the APOC–Willis 737-800 teardown, 2025]

Sale proceeds therefore arrive in layers, and the order matters for a working-capital business. Whole modules and life-limited parts with life can be sold to an engine shop or an exchange pool almost at once, because those buyers are ready-made (§36). Repaired rotatables arrive after repair-shop turnaround. The long tail of slow-moving airframe parts sells over years. A sourced figure for the total sell-through period, the months to recover most of the proceeds, was not captured

and is unknown U-D3-02; AerSale's inventory-turnover disclosure or a trade-press case study would answer it. [D3 §2.2, unknown U-D3-02]

The sequence by which an engine reaches a teardown is visible in one lessor's transactions. APOC leased a CFM56-5A to Condor for twelve months on a green-time lease, with the engine "expected to return to APOC's portfolio for teardown and part-out next year after the lease concludes". Green time, boxed in Part I §8 and owned by Part II §16, is the usable life left before the next mandatory shop visit; a green-time lease rents that life to an operator who will not restore the engine. [AviTrader, 28 Jul 2025; Aviation Business News, 2025; Part II §16] The same lessor acquired its first CFM56-5B for "up-cycling", its own term for buying an engine to tear down for material, in October 2021; has disassembled two CFM56-7Bs; bought four "young" 737 airframes from a large US legacy carrier for teardown; and in May 2026 bought a 15-year-old A320-200 from FTAI for disassembly at Tarmac Aerosave's Toulouse-Francal facility. [AviTrader, 7 Oct 2021; Aviation Business News, undated; ePlaneAI, 2026] That last item is a direct data point on FTAI as a seller of feedstock as well as a buyer (§35).

32.4 THE ENGINE ARITHMETIC THE TRADE PRESS PUBLISHES

One published source gives figures for a CFM56 teardown, and it is a consultancy blog, so it is labelled as such here and wherever it appears in this primer. [reconciliation E.1 rule 3] Safe Fly Aviation's 2026 report says an exhausted engine bought for \$0.8–1.2M can yield \$1.6–2.2M of used material, a net \$0.4–0.8M per engine. [Safe Fly Aviation, "CFM56 Engine Market Report 2026" (consultancy blog)]

RUN-OUT (EXHAUSTED) ENGINE

A run-out or exhausted engine is one with no usable time left before a shop visit: a life-limited part is at its limit, or the temperature margin is gone, or a hard inspection is due. It is the only kind of engine that becomes teardown feedstock, because an engine with life is worth several times its teardown yield as a whole (§32.5). Part II §16 carries what happens when green time runs out. [D3 §2.2–2.3; Part II §16]

WORKED EXAMPLE 5.1 — THE CONSULTANCY-BLOG ENGINE ARITHMETIC, AND WHAT IT LEAVES OUT

Inputs: purchase price of a run-out CFM56, \$0.8–1.2M; used-material yield, \$1.6–2.2M; stated net margin, \$0.4–0.8M (all sourced: Safe Fly Aviation, consultancy blog, 2026). The implied cost of teardown, repair, certification and selling is not stated by the source; the line below is arithmetic on the source's endpoints and is labelled derived.

low case: yield – purchase = $1.6 - 0.8 = \$0.8\text{M}$; stated net = $\$0.4\text{M}$;
 implied cost = $0.8 - 0.4 = \$0.4\text{M}$
 high case: yield – purchase = $2.2 - 1.2 = \$1.0\text{M}$; stated net = $\$0.8\text{M}$;
 implied cost = $1.0 - 0.8 = \$0.2\text{M}$
 implied cost as a share of yield: $0.4 \div 1.6 = 25\%$; $0.2 \div 2.2 = 9\%$

So in this source's framing, somewhere between a tenth and a quarter of gross used-material proceeds is consumed by the repair, test, tagging and sales effort, if the endpoints pair as written. Limits of the example: the source does not say its low and high cases pair that way; it does not give teardown labour or facility cost (unknown U-D3-09); and the yield is the sum of §32.2's five value pools without a split. Against the appraiser's half-life market values of \$5.7M for a -7B24 and \$6.4M for a -7B27, the whole teardown yield is a quarter to a third of a mid-life engine's worth, which is the point §32.5 makes. [Safe Fly Aviation, 2026 (consultancy blog); IBA (an appraiser), Engine Values 2025B, Sept 2025; D3 §2.2]

32.5 WHY ONLY RUN-OUT ENGINES BECOME FEEDSTOCK, AND THE OWNER'S CHOICE

The teardown yield has to be read against what the same engine is worth whole, and here the research holds a conflict it does not resolve. The appraiser IBA gives half-life market values for the second half of 2025 of \$5.7M for a CFM56-7B24 and \$6.4M for a -7B27, describes the -7B market as having "a distinct lack of availability" with values and lease rates above long-term trend, and in its first-half 2026 update says -5B and -7B values "have levelled off after approximately 18 months of increases". [IBA, Engine Values 2025B, Sept 2025; IBA, H1 2026 update] The consultancy blog gives a -7B "green-time value (new condition)" of \$2.8–3.4M, a label the research notes is internally inconsistent since a green-time engine is by definition not in new condition, and a teardown value of \$0.8–1.2M; a trade-press item has CFM56 values up "as much as 50% over the past two years". [Safe Fly Aviation, 2026 (consultancy blog); ePlaneAI, date not captured; reconciliation R-029] The figures sit on different bases, half-life against green-time against run-out, and they disagree on direction. Part I Exhibit 1.11 carries them side by side and this Part does the same.

WHY ONLY RUN-OUT ENGINES BECOME FEEDSTOCK

An engine near half-life, with a fresh restoration and life-limited parts around half their limits, is worth \$5.7–6.4M on the appraiser's figures. Torn down, the same engine would yield \$1.6–2.2M on the consultancy figures. No rational owner tears down a \$6M engine for \$2M of parts, so engines with life are sold or leased whole, and only engines at or near run-out reach the teardown market. The corollary runs the other way too: when half-life engines are scarce and expensive, as the appraiser describes for 2024–2026, owners of part-worn engines lease them for their remaining months rather than tear them down, and the feedstock pool shrinks further (§33). [IBA, Sept 2025 and H1 2026; Safe Fly Aviation, 2026 (consultancy blog); D3 §2.8 step 2, §5]

That choice, between renting out a part-worn engine for the months it has left and tearing it down now, has no name in the sources. This primer gives it one, because §36 needs it.

HALF-LIFE TRADE

The half-life trade is this primer's term, not the sources', for the trade in engines, modules and life-limited parts that are part-way through their lives and priced on the life they have left: a part-worn engine leased for its remaining months or torn down now; a module valued on the remaining cycles of the life-limited parts inside it; a used disk priced at its life fraction times list price less a discount (§34). The term collects what the dossier describes in three places without naming it. [D3 §2.3, §2.4–2.5, §2.7; registry section 1c]

WORKED EXAMPLE 5.2 — THE OWNER'S CHOICE AT THE END OF A LEASE: RENT THE GREEN TIME OR TEAR DOWN NOW

Inputs: a CFM56-7B with twelve months of green time (assumed term, matching the APOC–Condor twelve-month precedent on a -5A, sourced: AviTrader, 28 Jul 2025). Monthly rent \$45,000, the midpoint of the consultancy blog's \$42,000–48,000 green-time range (sourced: Safe Fly Aviation, 2026; midpoint is the dossier's illustrative choice). The alternative rent is the appraiser's about \$100,000 a month for a -7B spare engine in 2024, which may price a different kind of engine (sourced: IBA, Apr 2024; reconciliation R-028). Teardown yield \$1.6–2.2M (sourced: Safe Fly Aviation, 2026).

lease then tear down, at \$45,000: $12 \times 45,000 = \$540,000$ rent, then
 $\$1.6\text{--}2.2\text{M}$ of material = $\$2.14\text{--}2.74\text{M}$ over about a year plus sell-through
 lease then tear down, at \$100,000: $12 \times 100,000 = \$1,200,000$ rent, then
 $\$1.6\text{--}2.2\text{M} = \$2.8\text{--}3.4\text{M}$
 tear down now: $\$1.6\text{--}2.2\text{M}$, sooner, with no rent

The life-limited parts with twelve months of cycles left are worth slightly more in a teardown today than after another year of flying, since their remaining life is what a buyer pays for; the airfoils and cases are worth the same either way. Which path is chosen depends on the owner's cost of capital, the rent on offer, and whether an engine shop will pay more for the life now than a lessee will pay to burn it. One dossier divided the \$45,000 rent by the \$6.4M half-life value to get a lease rate factor of 0.70% a month, 8.4% a year, and noted that the calculation mixes a green-time rent with a half-life value; Part II §14 carries the lease rate factor and this primer states none as a single number. Limits of the example: the term is assumed; the two rents may describe different engines; and the yield is a consultancy range. [D3 §2.3; Safe Fly Aviation, 2026 (consultancy blog); IBA, Apr 2024; reconciliation R-028, R-030; E.1 rule 6]

32.6 THE FULL WORKED EXAMPLE: A 737-800 WITH TWO CFM56-7B ENGINES

The dossier assembled every input it could source for this example and left the rest open. The exhibit lists them, and the example that follows uses only the sourced ones, with each unsourced input labelled unknown rather than guessed. [D3 §2.8]

EXHIBIT 5.3 — INPUTS FOR A 737-800 PART-OUT: SOURCED FIGURES AND UNKNOWN

Input	Figure	Source, level and date	Status
Airframe purchase price, mid-life or older 737-800 bought for part-out	Not captured	None	Unknown U-D3-08
Engine purchase price, run-out CFM56, each	\$0.8–1.2M	Safe Fly Aviation, CFM56 Engine Market Report 2026 (consultancy blog)	Sourced
Engine used-material yield, run-out CFM56, each	\$1.6–2.2M	Safe Fly Aviation, 2026 (consultancy blog)	Sourced
Engine net margin after teardown, each	\$0.4–0.8M	Safe Fly Aviation, 2026 (consultancy blog)	Sourced
Half-life market value, CFM56-7B24 / -7B27 (ceiling if the engine has life)	\$5.7M / \$6.4M	IBA, Engine Values 2025B, Sept 2025 (appraiser)	Sourced
Green-time rent, CFM56-7B	\$42,000–48,000 a month; alternative about \$100,000 a month	Safe Fly Aviation, 2026 (consultancy blog); IBA, Apr 2024 (appraiser), basis may differ	Sourced, two readings (R-028)
Precedent green-time lease term	12 months (CFM56-5A, APOC to Condor)	AviTrader, 28 Jul 2025 (trade press)	Sourced
CFM56-5B life-limited parts, full life / half life, A320ceo	\$9,878,733 / \$4,939,366 (2023 dollars); per engine or per pair unstated	Mark Calver, Cargo Facts, Apr 2023 (conference presentation)	Sourced, basis unknown (U-D3-03)
CFM56-7B full life-limited-part set at list	\$5.7M (Jun 2025, scope not visible) or about \$4M (Nov 2019)	MyAirTrade (market-data site); Aircraft Value News (trade press)	Sourced, two readings (R-016)
Landing gear, fully overhauled set	"over \$1 million"	ISTAT Jetrader, Autumn 2020 (trade body magazine)	Sourced, 2020 figure

Input	Figure	Source, level and date	Status
Landing-gear overhaul cost	Up to \$400,000	ISTAT Jetrader, Autumn 2020	Sourced, 2020 figure
Airframe teardown duration	About 3 months	Asian Aviation / AirlinerGS, APOC–Willis 737-800 teardown, 2025 (trade press)	Sourced
Teardown labour and facility cost	Not captured	None	Unknown U-D3-09
APU, avionics, thrust-reverser and other rotatable values	Not captured	None	Unknown U-D3-10
Total sell-through period	Not captured	None	Unknown U-D3-02
Split of part-out value between engines and airframe	Not captured	None	Unknown U-D3-01

Source: D3 §2.8 inputs table; Safe Fly Aviation, 2026 (consultancy blog); IBA, Sept 2025 and Apr 2024; AviTrader, 28 Jul 2025; Mark Calver, Cargo Facts Session 3, Apr 2023; MyAirTrade, Jun 2025; Aircraft Value News, Nov 2019; ISTAT (the International Society of Transport Aircraft Trading) Jetrader, Autumn 2020, p.45; Asian Aviation and AirlinerGS, 2025. Reconciliation R-016, R-028, R-029.

WORKED EXAMPLE 5.3 — PART-OUT OF A 737-800 WITH TWO CFM56-7B ENGINES, INPUT BY INPUT

Step 1, the engines, both run-out. Purchase $2 \times \$0.8\text{--}1.2\text{M}$; yield $2 \times \$1.6\text{--}2.2\text{M}$; net margin per the source $2 \times \$0.4\text{--}0.8\text{M}$ (all sourced: Safe Fly Aviation, consultancy blog, 2026). These are one source's ranges applied to a pair, not two independent observations.

engine purchase:	$2 \times (0.8 \text{ to } 1.2) = \$1.6\text{--}2.4\text{M}$
engine yield:	$2 \times (1.6 \text{ to } 2.2) = \$3.2\text{--}4.4\text{M}$
engine net:	$2 \times (0.4 \text{ to } 0.8) = \$0.8\text{--}1.6\text{M}$

Step 2, engines with green time. If one engine has twelve months of green time (assumed), the owner can rent it at \$42,000–48,000 a month for \$504,000–576,000 before tearing it down (sourced rent: Safe Fly Aviation, 2026; at the appraiser's about \$100,000 a month, \$1.2M, sourced: IBA, Apr 2024). The purchase price of such an engine would be higher than a run-out engine's by roughly that rent stream, since that is what the seller gives up. If an engine is near half-life, its \$5.7–6.4M market value (sourced: IBA, Sept 2025) is several times its teardown yield and the owner sells or leases it whole; it never enters this example.

Step 3, the life-limited-part life inside the modules. For each engine the life-limited-part value at teardown is the sum over parts of $(\text{cycles remaining} \div \text{life limit}) \times \text{list price}$, then discounted to the

used price (\$34). The only current engine-level figure the research reached is the -5B figure: \$4.94M per engine at full life if Calver's \$9,878,733 is for the aircraft's two engines (sourced: Calver, Apr 2023; basis unknown, U-D3-03). An engine torn down with 30% average life remaining (assumed) carries:

life-proportional value: $0.30 \times 4.94\text{M} = \1.48M
 at the trader's 70–75% of list (sourced: Aircraft Commerce Issue 120):
 $1.48\text{M} \times 0.70 \text{ to } 0.75 = \$1.04\text{--}1.11\text{M}$
 on the -7B set of \$5.7M (sourced: MyAirTrade, Jun 2025, scope not
 visible): $0.30 \times 5.7\text{M} = \$1.71\text{M} \rightarrow \$1.20\text{--}1.28\text{M}$
 on the -7B set of \$4.0M (sourced: Aircraft Value News, Nov
 2019): $0.30 \times 4.0\text{M} = \$1.20\text{M} \rightarrow \$0.84\text{--}0.90\text{M}$

On any of the three set prices the life-limited parts at 30% life are most of the lower bound of the consultancy's \$1.6–2.2M yield, and the remainder is airfoils, cases and accessories. The three set prices are never mixed in one line (E.1 rule 5); the 2013 figure of \$1.7M for a -7B set is below the same publisher's 2008 figure and is not used (R-016).

Step 4, the airframe. The only sourced item is the landing gear: over \$1M for a fully overhauled set, or sold below half-life to a buyer who wants to avoid an overhaul costing up to \$400,000 (sourced: ISTAT Jetrader, Autumn 2020, a 2020 figure). The auxiliary power unit, avionics and the remaining rotatables are unknown (U-D3-10). The airframe purchase price is unknown (U-D3-08), so no airframe margin can be stated.

Step 5, timing. About three months to dismantle (sourced: APOC–Willis teardown, 2025); then repair-shop turnaround for repairable items; whole modules and life-limited parts with life placed with an engine shop or an exchange provider at once; the airframe tail sold over an unsourced period (U-D3-02).

What the example shows even with the gaps: on a run-out airframe the two engines alone return \$3.2–4.4M of used material on \$1.6–2.4M of engine cost, on one consultancy's figures; the single largest airframe item is about a quarter of one engine's yield; and the engine proceeds arrive fastest because the buyers are ready-made. What it cannot show: the airframe's purchase price and margin, the labour cost, the rotatable values other than the gear, and the months to full sell-through. [D3 §2.8; Safe Fly Aviation, 2026 (consultancy blog); IBA, Sept 2025 and Apr 2024; Calver, Apr 2023; MyAirTrade, Jun 2025; Aircraft Value News, Nov 2019; Aircraft Commerce Issue 120; ISTAT Jetrader, Autumn 2020; Asian Aviation, 2025; reconciliation R-016, R-028]

One more mechanism belongs here because it competes with teardown for the same airframes. A passenger-to-freighter (P2F) conversion, owned by Part II §15, gives a 20-plus-year-old airframe a second life; narrowbodies are converted at 21–22 years on average, and more than 260 737-800 freighters were in service as of mid-2025 with a pipeline exceeding 300. A conversion candidate still needs engines with life on them. So conversion is a source of demand for green-time engines and shop visits, not a source of teardown supply, and the IBA warning of September 2026 about

an "oversupply of narrowbody conversions" sits beside the pipeline figure without being reconciled in the research. [Cargo Facts and Acumen Aviation via the demand-context dossier (D9) §2.3; IBA, 6 Sep 2026, via the leasing dossier (D6); reconciliation R-034; Part II §15]

33. Supply: retirements, feedstock and the 2023–2026 shortage

Used material comes from aircraft and engines that leave service. Its supply is therefore a function of the retirement rate, lagged by the three months of teardown and the repair and certification period that follow. Part III §19 owns the retirement decision and the realised counts; this section carries what they mean for feedstock, and it presents every contested figure with both sources, as the reconciliation requires. [D3 §2.6; reconciliation E.1 rules 13 and 14]

33.1 THE RETIREMENT DECISION, AS IT BEARS ON FEEDSTOCK

The decision to retire a CFM56-powered aircraft is an economic one, and the trade press states a rule of thumb for it. A CFM56 needs a shop visit roughly every eight years, so the 24-year mark becomes a de facto retirement date because a third full overhaul is "deemed too expensive". Deliveries of the 737NG began in 1997 and ran at about 230 a year in 1998–2002, with A320ceo deliveries at a similar rate, so from 2022 the aircraft and engines of that period began entering their prime retirement window. [Aviation Week, "Retirement Uptick Will Restock Used Engine Parts", date not captured] That rule sits beside a different reading of the interval in the research: a mature CFM56 has a performance restoration at 10,000–15,000 cycles, which at 1,000–1,500 cycles a year is roughly seven to fifteen years, and a full overhaul with life-limited-part replacement at 20,000–25,000 cycles. The two are carried together and Part I §5 and §6 set out why they differ. [Leeham News, 8 Mar 2024; Aircraft Value News, undated; reconciliation R-008]

The rule of thumb has a feedstock consequence. If the third overhaul is the retirement trigger, then the engines reaching teardown are the ones whose life-limited parts are at or near their limits for the second time, and whose yield is therefore mostly airfoils, cases and accessories rather than life. Engines retired early, with life-limited parts part-way through a run, are the ones whose teardown carries life worth selling (§32.6 step 3). Which kind retires depends on the owner's alternatives, which is where the 2023–2026 shortage begins.

Part III §19 lists five inputs to the owner's decision: age and the maintenance cliff, lease economics, freighter conversion, part-out value, and the availability of a replacement. The fourth is this Part's subject. When an aircraft's value as a sum of parts exceeds its value flying, the owner parts it out; rising used-material prices raise part-out value and so, at the margin, encourage retirement. That is a stabilising feedback. The research records that it was not strong enough to lift retirements in 2024–2026, because the fifth input overrode it: none of the others

triggers a retirement if the airline cannot get a replacement aircraft. [D9 §2.3, the demand-context dossier (D9); Part III §19]

33.2 THE COUNTS AND THE RATES, WITH BOTH SOURCES

EXHIBIT 5.4 — RETIREMENTS AND FEEDSTOCK: EVERY FIGURE THE RESEARCH REACHED, WITH ITS BASIS, SOURCE LEVEL AND DATE

Quantity	Figure	Basis	Source and level	Date
Aircraft retirements, 2019 to 2022	502; 465; 333; 325	All commercial aircraft, Cirium Fleets Analyzer	Aerodynamic Advisory presentation to the IATA (the airlines' trade body) Maintenance Cost Conference (data provider via consultancy)	2023
Average annual part-outs, 2010–2019	Just under 540; peak 694 in 2013	Same	Same	2023
Retirement rate, 2024–2026 against 2010–2019	About 24% lower	Aircraft fleet	McKinsey (consultancy)	Date not captured (2024–25)
Normal retirement rate from 2028	About 2.7% of the fleet a year	Aircraft fleet	McKinsey	Same
A320ceo-family and 737NG retirements, next five years	1,190 and 795, together 35% of all retirements	Aircraft	Cirium / Ascend by Cirium (data provider)	Date not captured
737NG retired by 2040	About 72%	Aircraft	Ascend by Cirium	Date not captured
Types about half of retirements through 2025	A320ceo, 737NG, 777, A330 (an Airbus widebody)	Aircraft	Cirium via Informa Markets forecast page	Date not captured
Aircraft retirements, 2024	"Fewer than 200", similar to 2023, 70% below the 2020 peak	Scope not visible	IBA via AviTrader (appraiser)	24 Sep 2024

Quantity	Figure	Basis	Source and level	Date
Aircraft retirements, 2024	587	Scope not visible	Cirium Fleets Analyzer via Informa forecast page (data provider)	Date not captured
Passenger aircraft retired, share of active-plus-stored fleet	2.2% (2019); 0.4% (2024)	Passenger fleet	Aviation Week / Bain material (trade press, consultancy)	2024–25
CFM56 retirement planning assumption for 2026	3–4%, then 2–3%, then 1.5–2% of the installed base	CFM56 engines	GE Aerospace chief financial officer via Aviation Week (OEM via trade press)	2026
CFM56 retirements, 2025 actual; 2026 expected	About 1.5%, "broadly consistent with 2024"; about 2%	CFM56 engines	GE Q4 2025 results via LeeHam News (OEM via trade press)	22 Jan 2026
CFM56 retirements, first quarter 2026	"Below 1%"	CFM56 engines	GE Aerospace investor-relations page (OEM)	2026
CFM56 retirements	"Near 2% per year"	"Active" engines, about 14,200	Safe Fly Aviation (consultancy blog)	2026
CFM56-5B and -7B engines torn down per year, 2023–2025	Not captured	Engines	None	Unknown U-D3-17
Actual aircraft retirement counts, 2023–2025, from Cirium	Not captured (only 2019–2022 and the shortfall percentage)	Aircraft	None	Unknown U-D3-16

Source: IATA Maintenance Cost Conference, "Retirements, part-outs, dismantling", Aerodynamic Advisory (Murby), 2023; McKinsey, "What does the future hold for commercial-aviation maintenance?", date not captured; Cirium retirement-projection page and Informa Markets forecast page, dates not captured; AviTrader, 24 Sep 2024 (IBA data); Aviation Week, "Pockets of demand disrupt expected narrowbody retirement surge", and Bain, "Commercial aviation forecast"; Aviation Week, "CFM56 overhaul demand remains strong, GE Aerospace says", 2026; LeeHam News, 22 Jan 2026; GE Aerospace, "Recent events: your questions answered", 2026; Safe Fly Aviation, 2026. Reconciliation R-004, R-005; E.2 "Aircraft retirements" and "CFM56 retirement rate".

Three things in the exhibit need saying in words, because the figures do not reconcile and the primer does not reconcile them.

First, the two 2024 aircraft counts cannot both describe the same population. IBA's "fewer than 200" and Cirium's 587 differ by a factor of three, and neither snippet states its scope: passenger aircraft only, a size threshold, or year-to-date at the time of writing. They are carried side by side with "scope not visible". [AviTrader, 24 Sep 2024; Informa Markets forecast page; reconciliation R-005; E.1 rule 13]

Second, the aircraft-basis rates imply very different annual numbers. The reconciliation did the arithmetic on the dossier's figures. McKinsey's "24% below the 2010–2019 norm" applied to the Cirium average of just under 540 a year is about 410 aircraft a year. Cirium's projection that 1,190 A320ceo-family and 795 737NG aircraft will be 35% of retirements over five years implies about 5,670 retirements in total, about 1,130 a year. The two sit far apart, and the publication dates and scopes that might explain the gap were not captured. [reconciliation R-005, arithmetic on D3 §2.6]

McKinsey shortfall on the Cirium norm: $540 \times (1 - 0.24) \approx 410$ aircraft a year (derived)

Cirium five-year projection: $(1,190 + 795) \div 0.35 \approx 5,670$ aircraft over five years $\approx 1,130$ a year (derived)

Third, the engine-basis rates are GE's and they moved within one year. GE's planning assumption for 2026 went from 3–4% of the installed base to 2–3% to 1.5–2%; it reported about 1.5% retired in 2025, broadly the same as 2024; it expected about 2% in 2026; and first-quarter 2026 retirements ran "below 1%". [GE chief financial officer via Aviation Week, 2026; Leeham News, 22 Jan 2026; GE Aerospace investor-relations page, 2026; reconciliation R-004] The consultancy blog's "near 2% per year" is a fourth figure on a fourth basis, its own "active" count of about 14,200 engines, which one dossier read as the -7B fleet alone and another as all CFM56; both readings are recorded and neither is chosen. [Safe Fly Aviation, 2026 (consultancy blog); reconciliation R-002]

33.3 FEEDSTOCK ARITHMETIC ON THE INSTALLED BASE

The pool from which feedstock is drawn is the installed base Part I §2 boxed and Part III §18 owns: about 23,000 CFM56 engines in service on CFM's undated web page, about 24,000 in trade press citing September 2025, and more than 22,800 in Safran's end-2025 count; the consultancy blog's about 14,200 "active" and more than 19,000 "flying" are labelled consultancy figures and never the fleet figure. [CFM website, undated; Aviation Business News, Sept 2025; Safran, 13 Feb 2026; Safe Fly Aviation, 2026; reconciliation R-001; E.1 rule 14] Because no new -5B or -7B is built for an airframe, the base falls by exactly the number retired or converted each year, and that number is the gross supply of engine feedstock.

WORKED EXAMPLE 5.4 — ENGINES LEAVING THE BASE EACH YEAR, ON EACH PUBLISHED BASE AND RATE

Inputs: installed base about 24,000 (sourced: Aviation Business News, Sept 2025), about 23,000 (sourced: CFM website, undated) and more than 22,800 (sourced: Safran, 13 Feb 2026); CFM56 retirement rate about 1.5% (2025 actual) and about 2% (2026 expected) (sourced: GE via Leeham, 22 Jan 2026); consultancy base about 14,200 at "near 2%" (sourced: Safe Fly Aviation, 2026, consultancy blog). The alternative use of a retired core by FTAI's power business, 100 engines a year if one core powers one unit (assumed by the power dossier (D8), Part IX §74), is shown as a share.

on 24,000: $\times 1.5\% = 360$ engines a year; $\times 2\% = 480$
 on 23,000: $\times 1.5\% = 345$; $\times 2\% = 460$
 on 22,800: $\times 1.5\% = 342$; $\times 2\% = 456$
 on the consultancy base, $14,200 \times 2\% = 284$
 first quarter 2026 "below 1%" annualised on 24,000: fewer than 240
 100 cores a year for power as a share: $100 \div 480 = 21\%$; $100 \div 360 =$
 28% ; $100 \div 284 = 35\%$

So the OEM-level bases and GE's realised rate put gross engine retirements at roughly 340 to 480 a year, and the consultancy base and rate at about 284. Not every retired engine is torn down: some are sold whole to another operator, some are leased for their green time first (§32.5), and from 2026 some are earmarked as power-generation cores (Part IX). The number actually torn down per year was not captured (U-D3-17). Limits of the example: the bases may or may not include spare and stored engines (U-D9-03); the rates are a planning assumption, a realised figure and a consultancy figure on different denominators; and one core per power unit is an assumption. The two reconciliation rules that govern the presentation are that every base and every rate carries its source and that the consultancy figures are labelled. [reconciliation R-001, R-002, R-004; C.8 item 1; E.1 rules 3, 13, 14; D8 §2.10]

33.4 THE 2023–2026 SHORTAGE: WHAT THE SOURCES DESCRIBE

The mechanism, as the sources describe it, has six steps. Airlines kept CFM56-powered aircraft flying longer because new narrowbody deliveries slipped and new-generation engines spent time off wing. The expected post-2022 retirement wave therefore did not arrive. Retirements stayed well below the 2010–2019 norm. The engines that would have been feedstock were instead leased as green-time engines or overhauled. Teardown companies competed for a smaller pool. And the price of everything in the pool rose. [D3 §2.6; D9 §2.5–2.8]

Part III owns the first step and this Part takes its headline figures from it. Cirium data cited in trade press put 835 aircraft powered by Pratt & Whitney's geared turbofan (GTF) grounded or stored in late 2025. CFM's LEAP (the CFM56's successor) was removed early at 2,000–6,000 cycles against a mature CFM56's 10,000–15,000-cycle interval, and its 737 durability kit was certified only in July 2026. Boeing's 737 MAX (the 737NG's successor) was capped by the FAA

at 38 a month after January 2024, raised to 42 in October 2025 and reviewed for 47 in May 2026. Airbus delivered 793 aircraft in 2025 against an 820 target. And 75–90% of scheduled lease expiries were being extended, with only about a dozen mid-life 737NGs and fewer than 25 A320neos available to rent worldwide at the time of that survey. [The Flying Engineer citing Cirium, 2026; Leeham News, 8 Mar 2024; CFM release, 18 Jul 2026; TravelPulse, Oct 2025; CAPA (the Centre for Aviation), May 2026; Air Data News, 2026; Bloomberg via AJOT (American Journal of Transportation), date not visible; Part III §20–22]

The consequences in the figures the research captured are these. CFM56 values rose "as much as 50%" in two years on one trade-press account. [ePlaneAI, date not captured] CFM56-7B green-time rents in 2026 stood about 34% above 2019 on the consultancy blog, and the appraiser's spare-engine rent rose from about \$75,000 to about \$100,000 a month from 2019 to 2024, also about a third; the levels differ and the percentage change agrees, which the reconciliation records as one of the few independent corroborations in this area. [Safe Fly Aviation, 2026 (consultancy blog); IBA, Apr 2024; reconciliation B-5, R-028] IBA called it "a lessors' market" in April 2024 as engine lease rates and market values escalated. [IBA via AviTrader, 25 Apr 2024] Aviation Week headlined "a seller's market for used parts" in a column on MRO (maintenance, repair and overhaul). [Aviation Week, "MRO Memo: A Seller's Market For Used Parts", date not captured] AerSale's management described "a hypercompetitive feedstock environment, marked by increased scrutiny and pressure on pricing as more companies enter the teardown sector", and AerSale's whole-asset sales fell from 20 engines and one aircraft in 2024 to 13 engines in 2025. [AerSale, as relayed in search results; AerSale release, 5 Mar 2026] The three descriptions agree that asset owners had the upper hand and teardown buyers did not; they differ only in vantage point, and the dossier recorded that as its sixth tension rather than a contradiction. [D3 tension 6]

Whether the parts constraint was easing by 2026 is contested, and both sides are carried. GE's FY2025 annual report says material input from priority suppliers rose more than 40% and turnaround improved more than 10% in the fourth quarter of 2025. Oliver Wyman's April 2026 survey has two-thirds of maintenance operators unable to find enough technicians and labour-rate inflation at 5.5–6.0%, and a 2025 trade-press report says used-material supply "remains thin". [GE Aerospace 10-K FY2025; Oliver Wyman, Apr 2026; AVM (Aviation Maintenance magazine), 2025; reconciliation R-020]

THE SAME ENGINE ON BOTH SIDES OF THE AFTERMARKET

Every CFM56 that is not retired keeps flying, keeps accumulating cycles, and will need a shop visit (Part IV). Every CFM56 that is not retired is also one that is not torn down, so its life-limited parts, airfoils and modules do not reach the used-material market. The driver that raises demand for used parts therefore lowers their supply at the same moment, with the same engine. Part I §2.3 made the point from the fleet side and Part III §24 turns it into translation-table rows; this Part is where the supply side of it is counted. [D9 §5; Part I §2.3; D3 §2.6]

33.5 THE FORECAST THAT POINTED THE OTHER WAY

In July 2020 Oliver Wyman published a piece titled "A tsunami of used aircraft parts", forecasting a glut of used material from pandemic retirements. The research captured only the title, so none of its figures is reported, and whether Oliver Wyman has since updated the forecast is unknown U-D3-19. [Oliver Wyman, Jul 2020, title only; D3, unknown U-D3-19] The outturn pointed the other way on every count the research reached: Cirium's counts fell every year from 502 in 2019 to 325 in 2022; McKinsey reports 2024–2026 retirements about 24% below the 2010–2019 norm with a return to about 2.7% of the fleet a year only from 2028; GE's realised CFM56 rate was about 1.5% in 2025 against a plan that had started at 3–4%; and AerSale, IBA and Aviation Week all describe scarcity. [IATA Maintenance Cost Conference, 2023; McKinsey, date not captured; LeeHam News, 22 Jan 2026; D3 tension 2; reconciliation B-6] The forecast and the outturn are both carried; the forecast's own numbers are not, because they were not read.

Two Aviation Week headlines in the research point in different directions within the shortage period: "Used Parts Market Shows Signs Of Stabilizing" and "North America Sees Upturn In USM Market". Their dates and bodies were not captured, so the sequence cannot be fixed; IBA's "levelled off" of the first half of 2026 and the consultancy blog's rents "up 15–20% in recent months" sit in the same unresolved tension. [Aviation Week, two titles, dates not captured; IBA, H1 2026; Safe Fly Aviation, 2026; D3 tension 7]

33.6 WHAT GE'S USED-MATERIAL PROGRAMMES MEAN FOR INDEPENDENTS' FEEDSTOCK

The manufacturer is also a used-material trader, and by its own statement the largest. GE Aerospace said in 2026 that "GE and CFM are the largest used serviceable material provider for their products". [GE Aerospace, 2026, via Aviation Week and GE investor materials; D2 §2.2, the shop-visit market dossier (D2)] Part IV §28 owns the programmes; what belongs here is where they touch feedstock.

Two of GE's four branded TrueChoice offerings carry used material. TrueChoice Material sells "high-quality new and used OEM parts, advanced repairs and technology upgrades" to airlines and to third-party shops; Ryanair's long-term materials agreement covering "their entire fleet of ~2,000 CFM56 and LEAP engines" is a contract of that kind. TrueChoice Transitions offers "green time leases, exchanges and material buy-back, plus custom worksopes with shorter builds and maximum used material" to operators changing their ownership horizon. [GE Aerospace press releases, 2019–2025; GE Aerospace 10-K FY2025; D2 §2.3]

For an independent teardown company the programmes mean three things, each a description rather than a judgment. First, the manufacturer competes for retired engines: material buy-back under Transitions takes engines at the end of an operator's horizon into the OEM's own pool before a trader can bid. Second, the manufacturer sells the material those engines yield through Material, into the same shops the independents supply, and it can bundle it with new parts and

repairs, which an independent cannot. Third, a used part sold by the manufacturer carries the manufacturer's release, which matters to a buyer who vets suppliers as §31.2 described. A GA Telesis purchase of \$25M of "OEM-aligned" used engine inventory in October 2025 shows OEM-sanctioned used-material channels running beside independent ones. [D2 §2.3; GA Telesis release, Oct 2025; D3 §3] The name and scale of GE's used-material arm, and its CFM56 role, were not captured; the only hit was a TrueEngine presentation whose content was not read, and the gap is clustered with Part VI's open question on GE's positioning against alternative parts (K-16). [D3, unknown U-D3-15; reconciliation K-16]

34. Pricing versus new OEM parts

A used part has no price list of its own. It is priced as a percentage of the manufacturer's catalogue list price for the same part number, and that one fact explains most of what follows: the used price inherits the manufacturer's annual escalation, the discount narrows when the manufacturer cannot deliver, and it widens when retirements flood the market. [D3 §2.5]

USM PRICING VERSUS NEW (THE USM DISCOUNT TO CLP)

Used serviceable material is priced as a percentage of the OEM's catalogue list price (CLP) for the same part: on average 60–80% of a new part's price; 70–75% of list when a specialist engine trader sells repaired material to an airline; up to 85% of list for high-demand parts early in a platform's life; and above list when the manufacturer cannot deliver. Because the base is the catalogue, the used price rises with every catalogue revision even if the discount does not move. [Aviation Week, "MRO Memo: A Seller's Market For Used Parts", date not captured; Aircraft Commerce Issue 120, date not captured; D3 §2.5]

34.1 THE PUBLISHED FIGURES AND THEIR TENSION

EXHIBIT 5.5 — USED-PART PRICES RELATIVE TO NEW, AS PUBLISHED, WITH BASIS, SOURCE LEVEL AND DATE

Figure	What it describes	Source and level	Date
60–80% of a new part's price	The average used part, all classes	Aviation Week, "MRO Memo: A Seller's Market For Used Parts" (trade press)	Date not captured
Up to 85% of catalogue list price	High-demand used parts early in a platform's life cycle	Aviation Week, same	Date not captured
"More than the catalog list price"	Used parts when the OEM has supply-chain problems	Aviation Week, same	Date not captured

Figure	What it describes	Source and level	Date
70–75% of catalogue list price	Repaired engine used material sold by a specialist trader to an airline	Aircraft Commerce Issue 120 (trade press)	Date not captured
Upward trend "most evident on CFM56 and V2500 platforms"	Driven by rising demand, falling supply of new parts and used parts, and rising repair costs	Aviation Week, "Magnetic expects CFM56 market challenges", 2024 (trade press)	2024
CFM raised its catalogue list price	"A notable shift in pricing strategy for this platform"	Same	Aug 2023
Residual value assigned to "alternative materials"	60–70% of the industry assigns 25% or less	Safe Fly Aviation teardown report (consultancy blog)	2026
Prices of used life-limited-part packages with stated cycles remaining	Not shown in the listings reached	None	Unknown (U-D1-04)

Source: Aviation Week, "MRO Memo: A Seller's Market For Used Parts", date not captured; Aircraft Commerce, Issue 120 maintenance article, date not captured; Aviation Week, "Magnetic expects CFM56 market challenges, opportunities in 2024", 2024; Safe Fly Aviation, "Aircraft Teardown Market Growth", 2026 (consultancy blog); the engine dossier (D1), unknown U-D1-04. Reconciliation R-022.

The figures are consistent as a range and not as a single number. Aviation Week's 60–80% is a general average across every class of used part; Aircraft Commerce's 70–75% is a specific price, a trader selling repaired engine material to an airline; the 85% and above-list cases are the ends of the same spectrum. Which applies depends on the part, the platform and the manufacturer's supply at the time, and the primer carries all four rather than one. [reconciliation R-022; D3 tension 5] The V2500 named in the exhibit is the alternative A320ceo engine of Part I §10.

34.2 LIFE-LIMITED PARTS: LIFE FRACTION FIRST, THEN DISCOUNT

For most used parts the percentage is applied to the list price directly. For a life-limited part there is an extra step, because what the buyer is purchasing is the remaining cycles. The life-proportional value is the fraction of life remaining times the list price of the new part; the used price is that figure less the trader's discount. Part I §4.4 stated the rule from the engine side; here it is from the trader's. [D3 §2.4–2.5; Part I §4.4]

The one current list price the research reached for a CFM56-7B life-limited part is a consultancy figure: a high-pressure turbine disk at \$420,000 new, with its 20,000-cycle limit. The current list price of representative -7B life-limited parts and of a full set was otherwise not captured and is unknown U-D3-06. [Safe Fly Aviation, 2026 (consultancy blog), via D2 §4.4; D3, unknown U-D3-06]

disk with 70% of life left: life-proportional value = $0.70 \times 420,000 = \$294,000$
 at the trader's 70–75% of list: $294,000 \times 0.70$ to $0.75 = \$206,000$ – $221,000$,
 or 49–53% of the new price
 buyer facing a one-year OEM lead time: may pay $0.70 \times 420,000 = \$294,000$ or
 more, the "above list" case in life-adjusted terms

That last line is how the trade-press statement that used parts can sell "for more than the catalog list price" should be read for a life-limited part. The buyer is not paying more than list for a new disk. It is paying the full life-proportional value, or above it, for a used disk it can have now, because the alternative is a grounded engine for the manufacturer's lead time. The mechanism matters more than the specific disk: the discount to list narrows when the manufacturer's own supply is constrained, and widens when retirements flood the market with used parts. [Aviation Week, date not captured; D3 §2.5]

34.3 HOW A USED PART INHERITS THE MANUFACTURER'S ESCALATION

Part I §4.3 boxed escalation and Part IV §28 owns it. Two readings sit in the research and both are carried. The first covers all spare parts: new spare-parts prices rising "around 10% per year", a double-digit increase on 1 November 2022, a high-single-digit increase in August 2023, and Safran's chief executive describing a rule of pricing "above inflation by 3 to 4 points ... at least for some years". [Aviation Week, "Parts Price Hikes Help Boost Safran, GE Aftermarket Sales", late 2023] The second is specific to life-limited parts: a full -7B set compounded at about 7% a year from 2008 to 2025 on the published set prices, a single step of about 12% reported by Ishka without a year, and -7B parts generally up 20–30% in 2023 after CFM's August 2023 catalogue revision. [Part I worked example 1.3; Ishka, undated; Magnetic Group via Aviation Week, 2024; reconciliation R-018] The years and amounts of each catalogue step between 2019 and 2026 were not retrieved (cluster K-03).

Because the used price is a percentage of the catalogue, every one of those steps moves it. The worked example shows the arithmetic over three years on both escalation readings and on both discount readings.

WORKED EXAMPLE 5.5 — ONE PART AT LIST AND AS USED MATERIAL OVER THREE ESCALATION YEARS

Inputs: a CFM56-7B high-pressure turbine disk at \$420,000 new in year 0 (sourced: Safe Fly Aviation, 2026, consultancy blog; the only current -7B life-limited-part price reached). Escalation case A, 7% a year (derived in Part I from the 2008–2025 set prices). Escalation case B, 10% a year (sourced: Aviation Week, late 2023, all spares). Used discount reading one, 70–75% of list (sourced: Aircraft Commerce Issue 120); reading two, 60–80% (sourced: Aviation Week). The example prices the disk as an ordinary used part, that is, before any life-fraction adjustment, to isolate the escalation effect; §34.2 adds the life step.

case A, 7% a year: list 420,000 → 449,400 → 480,858 → 514,518
 used at 70–75%: 294,000–315,000 → 314,580–337,050 → 336,601–
 360,644 → 360,163–385,889
 case B, 10% a year: list 420,000 → 462,000 → 508,200 → 559,020
 used at 70–75%: 294,000–315,000 → 323,400–346,500 → 355,740–
 381,150 → 391,314–419,265
 used at 60–80% of the year-3 list: case A 308,711–411,614; case B
 335,412–447,216
 dollar saving from buying used at 70–75%, year 0 → year 3: case A
 \$105,000–126,000 → \$128,630–154,355; case B → \$139,755–167,706

Three things follow. The used price in dollars rises every year even though the discount is unchanged, by exactly the manufacturer's escalation rate: a trader holding a disk in stock sees its selling price rise with the catalogue. The dollar saving from buying used also grows with escalation, because it is a fixed share of a rising list. And a 20–30% single-year step of the kind reported for -7B parts in 2023 lifts the used price by the same 20–30% in the year it lands. The one thing escalation does not do is close the gap between used and new; only the manufacturer's delivery constraint does that, by narrowing the discount toward 85% of list and above. Limits of the example: the disk price is a consultancy figure; the 7% rate is derived from set prices whose scope differs; the catalogue steps for 2024–2026 are unknown; and the used readings describe different transaction types (R-022). [Safe Fly Aviation, 2026 (consultancy blog); Aviation Week, late 2023; Aircraft Commerce Issue 120; Part I worked example 1.3; reconciliation R-018, R-022; K-03]

HOW THE CATALOGUE SETS THE USED PRICE

The manufacturer never prices a used part, and it sets the price of every used part. It does so because the trade quotes used material as a share of list, and the manufacturer alone sets list and revises it upward on a schedule. A used-material trader therefore carries two exposures it does not control: the manufacturer's escalation, which lifts the value of its stock, and the manufacturer's delivery performance, which sets how close to list it can sell. A third party to the same line in the parts bill is the alternative-manufacturer part of Part VI, which is priced against the same catalogue; the two compete for the same slot, and an engine with alternative parts carries less of the list-price-based maintenance adjustment that appraisers apply, which Aircraft Value News records as undermining full-life adjustments (Part I §8). [D3 §2.5, §5; D4 §2.1; Aircraft Value News, undated]

35. The players

The used-material trade has four kinds of participant: independent traders that buy feedstock and sell parts, lessors that route their own retired assets into captive material arms, the manufacturer's own used-material business, and FTAI, which does something different with the same feedstock. This section gives one paragraph to each named company, with the scale its filings or releases disclose and nothing more, and ends with the market-size estimates. Two listed companies,

AerSale and AAR, are also peers in Part XI §87, which carries their full filing treatment. [D3 §3; D11 §3.5, §3.7, the peers dossier (D11)]

35.1 ONE PARAGRAPH EACH

AerSale. AerSale is the cleanest listed example of a teardown-and-used-material business and is listed on Nasdaq. Its Asset Management Solutions (AMS) segment "extract[s] value from strategic asset acquisitions either as whole assets or by disassembling for used serviceable material"; its TechOps segment is maintenance and engineered products. For 2025 it reported revenue of \$335.3M, down 2.8% from \$345.1M; AMS revenue of \$211.6M against \$215.5M; TechOps revenue of \$123.7M against \$129.6M; used-material revenue of \$137.6M, made up of \$120.1M in AMS and \$17.6M in TechOps (the components sum to \$137.7M, a rounding difference); and flight-equipment sales of \$56.4M from 13 engines, against \$110.1M from 20 engines and one aircraft in 2024. Revenue excluding flight-equipment sales rose 18.7%. [AerSale release, 5 Mar 2026; AerSale 10-K FY2025] Its AMS gross margin was 35.0% in 2025 against 38.3% in 2024, both derived from the filing's gross-profit and sales figures, with the decline attributed to "lower margin on USM sales resulting from changes in the product mix"; engine revenue rose on higher used-material sales in the PW4000 (a Pratt & Whitney widebody engine) and CF6-80 lines, which are not CFM56. Fourth-quarter 2025 feedstock acquisitions were \$15.4M; a full-year purchase total was not returned. [AerSale 10-K FY2025, via D11 §3.5 and §2.4; reconciliation B-8]

WHOLE-ASSET SALE (FLIGHT-EQUIPMENT SALE)

A whole-asset sale is the sale of a complete aircraft or engine from a trader's inventory rather than its disassembly into parts; AerSale's filings call the line "flight equipment sales". AerSale sold 13 engines this way in 2025 against 20 engines and one aircraft in 2024. The choice between selling whole and tearing down is the one §32.5 described from the owner's side. [AerSale release, 5 Mar 2026; D3 §3]

ADJUSTED EBITDA (EBITDA, ADJUSTED)

EBITDA is earnings before interest, taxes, depreciation and amortization. Adjusted EBITDA is that figure further adjusted by the company for items it treats as non-recurring or non-cash, and it is a company-defined profit measure; no two companies' definitions are the same. AerSale reported Adjusted EBITDA of \$46.1M for 2025, 13.8% of revenue, against \$33.4M and 9.7% in 2024. Part X §80 carries FTAI's definition and reconciliation. [AerSale release, 5 Mar 2026; Part X §80]

GA Telesis. GA Telesis (GA) is a privately held trader, lessor and maintenance provider whose Flight Solutions Group buys engines for disassembly and sells the material. In January 2025 it bought eight PW4000 engines from a major US airline for disassembly; in November 2025 it disassembled eight more, one CFM56-5B, two V2500-A5, two PW4000 and three CF6-80C2;

and in October 2025 it bought an additional \$25M of "OEM-aligned" used engine inventory. It signed a CFM56-5B and -7B material agreement with Aero Engine Solutions in January 2018. Its revenue is not disclosed. [GA Telesis releases, Jan and Nov 2025; Aviation Business News, undated; AviTrader, 31 Jan 2018]

VAS Aero Services. VAS Aero Services (VAS) is a private teardown-management and used-material redistribution company that also handles large aircraft: Airbus appointed it to oversee the dismantling and material redistribution of three retiring A380 (Airbus's double-deck widebody) aircraft. Its revenue is not disclosed; a trade-press item quoting it refers to a "\$29 billion MRO market", MRO (maintenance, repair and overhaul) being the industry's generic term, without defining the scope. [ePlaneAI, undated; D3 §3, §4.5]

AAR. AAR is listed on the NYSE and is the scale reference for a parts business that mixes distribution of new manufacturer parts with used material. Its Parts Supply segment, "distribution of new OEM parts and sales and leasing of USM", "accounted for approximately 45% of AAR's sales in fiscal 2026", the year to 31 May 2026. Consolidated sales were \$3,308.0M, derived from \$2,384.1M of commercial and \$923.9M of government sales, which puts Parts Supply at about \$1,489M, also derived. Segment sales rose \$388.1M year on year; new-parts distribution grew 23.9% and used-material activities 32.1%; whole-asset sales in parts trading rose \$27.8M in the first quarter of fiscal 2026; and the segment absorbed the September 2025 acquisition of American Distributors Holding Co. (ADI) for \$137.1M. The split of Parts Supply between distribution and used material, and the segment's operating income, were not returned. [AAR 10-K FY2026; AAR 10-Q for the quarter ended 31 Aug 2025; D11 §3.7, unknown U-D11-14; D3, unknown U-D3-11]

DISTRIBUTION (NEW PARTS)

Distribution is the resale of new manufacturer parts under a distribution agreement: the distributor holds inventory and earns the margin between the manufacturer's transfer price and its selling price. Its margins are structurally thinner than those on used or alternative parts, because the manufacturer sets the cost of goods. AAR's Parts Supply combines it with used-material sales and leasing; FTAI has no distribution business. Part XI §87 carries AAR as a peer. [D11 §2.4, §3.7]

AJW Group. AJW Group (AJW) is a private component-support and used-material company and the author, with AviTrader, of the August 2025 article on used-material airworthiness on which §31 rests. Its scale was not captured and is unknown U-D3-13. [AJW Group via AviTrader, Aug 2025; D3, unknown U-D3-13]

Willis Aero. Willis Aero is the parts subsidiary of Willis Lease Finance, the only US-listed company whose primary business is engine leasing (Nasdaq-listed; Part XI §87). It sells engine parts and materials "through the acquisition or consignment of aircraft and engines". Engines removed from the lease portfolio for part-out are carried in "equipment held for sale", and the

investment is "recovered through the sale of spare parts". It partnered with APOC on a 737-800 teardown. The parent's 2025 figures give its feedstock scale: 363 engines and 20 aircraft in a lease portfolio of \$2,801.7M of equipment held for lease, 116 further engines managed for third parties, average utilisation of 84.9%, revenue of \$730.2M and lease rent of \$291.6M. Willis Aero's own spare-parts revenue was not returned and is clustered as an open question with Part XI (K-15). [Willis Lease Finance 10-K FY2025; Asian Aviation, 2025; D11 §3.1; reconciliation K-15]

Setna iO. Setna iO is a private used-material trader whose reported teardown activity reaches the new generation: it acquired a 2019-vintage A320neo airframe and two sets of quick-engine-change (QEC) components, the mounts, plumbing and accessories that dress a bare engine for installation, for the geared-turbofan engine. Its revenue is not disclosed. [AviTrader, undated item; D3 §3]

GE's used-material arm. The manufacturer's own used-material business sits inside GE Aerospace's TrueChoice Material offering, described in §33.6, and GE states that "GE and CFM are the largest used serviceable material provider for their products". Its name as a business unit, its scale and its CFM56 role were not captured (U-D3-15; K-16). [GE Aerospace, 2026; D2 §2.3; D3, unknown U-D3-15]

AerCap Materials. AerCap Materials is the part-out and used-material arm of AerCap, the largest aircraft lessor, which files a Form 20-F as a foreign private issuer. It was formed when AerCap acquired GECAS (GE Capital Aviation Services) in late 2021, which had a parts business active in teardowns. It dismantles at Greenwood-Leflore County Airport in Mississippi and distributes from Memphis, Tennessee. Its 2023 factsheet records more than 300 aircraft dismantled and more than 15,000 unique parts; its first-quarter 2026 factsheet records more than 250,000 items stocked and more than 850 customers, with dismantling offered "with tailored workscopes, competitive pricing and accelerated turnaround time". Its revenue is not disclosed in the filing text returned. The parent's feedstock scale: 470 owned engines at 31 December 2025, stated as 9% of its fleet, or more than 1,200 engines including those owned and managed by its Shannon Engine Support (SES) joint venture with Safran at 30 September 2025; the two counts are on different bases and both are carried. [AerCap Materials factsheets, 2023 and 1Q 2026; Aviation Week, "Teardown providers expand business", date not captured; AerCap 20-F FY2025; AerCap 6-K, Q3 2025; D11 §3.2; reconciliation R-033]

APOC Aviation. APOC is a private engine and airframe trader and lessor that leases green time and tears down. Its transactions in the research: the first CFM56-5B bought for up-cycling in October 2021; two CFM56-7Bs disassembled; four young 737 airframes bought from a US legacy carrier for teardown; the 737-800 teardown with Willis Lease; the twelve-month CFM56-5A green-time lease to Condor with teardown to follow; and the 15-year-old A320-200 bought from FTAI for disassembly at Tarmac Aerosave's Toulouse-Francalaz site in May 2026.

Tarmac Aerosave is a dismantling and storage company owned by Airbus, Safran and Suez. [AviTrader, 7 Oct 2021 and 28 Jul 2025; Aviation Business News, undated; Asian Aviation, 2025; ePlaneAI, 2026]

Unical Aviation. Unical is a private used-material trader on which the dossier's own searches returned nothing; the orchestrator's later searches established one fact, which is that Unical was FTAI's joint-venture partner when the assets of iAero Thrust were acquired on 4 January 2023 to form QuickTurn in Miami, and that FTAI bought Unical's remaining interest on 1 December 2023 for \$30.3M in cash. Unical's own scale remains unknown (U-D3-14, partly resolved). [GlobeNewswire, 1 Dec 2023; FTAI 10-K FY2025; orchestrator notes; reconciliation D.3]

FTAI's own sourcing. FTAI is both a buyer and a seller of feedstock, and it does something with it that the traders above do not: it rebuilds the material into modules. Its FY2025 annual report says that engines are "transferred to inventory when determined to be unserviceable for leasing purposes with the intent to tear down the asset into salvageable components that would be manufactured into new modules and parts and sold"; that the Aerospace Products business "focuses on the manufacturing and sale of new assets through the use of inventory purchased from third parties and salvaged modules and parts from generally unserviceable engines"; and that "serviceable used modules and parts are sold through an exclusive partnership responsible for the teardown, repair, marketing and sales of parts from the CFM56 engine pool". [FTAI 10-K FY2025] The filing excerpt does not name the partner. A 2025 PRNewswire release is titled "AAR and FTAI Aviation extend their exclusive Serviceable Engine Products agreement, providing CFM56 engine material to the global aviation aftermarket through 2030"; the title was located by two dossiers and the body was read by neither, so the primer records that the filing's description and the partner's release title are consistent and that the agreement's terms are unknown (U-D3-07). [PRNewswire, 2025, title only; reconciliation B-9, C.3 item 1] FTAI's captive feedstock flows are described in the Parts that own them: a leasing fleet of 243 engines at 31 December 2025 (Part VIII §58); the Strategic Capital Initiative (SCI) vehicles whose engines are powered through exchanges (Part VIII §61); the ten off-lease 737-700s bought from WestJet in September 2026 for their CFM56-7B engines and modules (Part VIII §67); and, from 2026, the power business that consumes CFM56 cores (Part IX §74). On the selling side, the A320-200 sold to APOC for disassembly shows FTAI releasing an airframe it did not need. Part VII §46 owns the Module Factory into which the engine feedstock goes. [FTAI 10-K FY2025, via D6; FTAI release, 28 Sep 2026, via D6; ePlaneAI, 2026; D5 §2.2, the Aerospace Products dossier (D5)]

EXHIBIT 5.6 — THE PLAYERS: WHAT EACH DOES IN USED MATERIAL AND THE SCALE EACH DISCLOSES

Company	Status	Role in used material	Scale disclosed (source, date)
AerSale	Listed (Nasdaq)	Buys mid- and end-of-life aircraft and engines; leases, sells whole or tears down; maintenance (TechOps)	Revenue \$335.3M; AMS \$211.6M at 35.0% gross margin (derived); used-material revenue \$137.6M; 13 engines sold whole; Adjusted EBITDA \$46.1M (release, 5 Mar 2026; 10-K FY2025)
AAR	Listed (NYSE)	Distribution of new parts plus used-material sales and leasing; whole-asset trading	Parts Supply about 45% of sales, about \$1,489M (derived) of \$3,308.0M (derived); used-material activity +32.1% (10-K FY2026)
Willis Lease / Willis Aero	Listed (Nasdaq)	Engine lessor; captive parts arm by acquisition or consignment	363 engines, 20 aircraft, \$2,801.7M equipment held for lease, 84.9% utilisation; parts revenue not returned (10-K FY2025)
AerCap / AerCap Materials	Listed (NYSE; 20-F filer)	Largest aircraft lessor; captive teardown arm since the GECAS acquisition, 2021	300+ aircraft dismantled, 15,000+ unique parts (2023); 250,000+ items, 850+ customers (1Q 2026); 470 owned engines, >1,200 including SES (20-F; 6-K)
GE Aerospace	Listed (NYSE)	OEM; "largest used serviceable material provider for their products"; TrueChoice Material and Transitions	Name and scale of the used-material arm not captured (GE Aerospace, 2026; U-D3-15)
FTAI Aviation	Listed (Nasdaq)	Tears down engines unserviceable for leasing; rebuilds into modules; sells used modules and parts through an exclusive partnership; also sells airframes for teardown	Segment figures in Part VII; partner unnamed in the filing; AAR's release (27 Mar 2025) names AAR as exclusive provider for FTAI's pool of over 450 CFM56 engines, handling teardown, repair, marketing and sales, through 2030 (10-K FY2025; PRNewswire, 2025)
GA Telesis	Private	Buys engines for disassembly; "OEM-aligned" used inventory; leasing; maintenance	16 engines disassembled in two 2025 batches; \$25M inventory purchase, Oct 2025; revenue not disclosed (releases)
VAS Aero Services	Private	Teardown management and redistribution, including widebodies	Three A380 teardowns for Airbus; revenue not disclosed (ePlaneAI)

Company	Status	Role in used material	Scale disclosed (source, date)
AJW Group	Private	Component support; used material	Not captured (U-D3-13)
Setna iO	Private	Used-material trader; new-generation teardown	One A320neo airframe and two GTF QEC sets; revenue not disclosed (AviTrader)
APOC Aviation	Private	Trader and lessor; green-time leases; teardown ("up-cycling")	Transactions listed in §32.3 and §35.1; revenue not disclosed
Unical Aviation	Private	Used-material trader; FTAI's former QuickTurn partner	\$30.3M paid by FTAI for its remaining interest, 1 Dec 2023; own scale unknown (U-D3-14)
Tarmac Aerosave	Private (Airbus, Safran, Suez)	Dismantling and storage facilities	Site of the FTAI-sourced A320-200 teardown (ePlaneAI)

Source: AerSale release, 5 Mar 2026, and 10-K FY2025; AAR 10-K FY2026; Willis Lease Finance 10-K FY2025; AerCap 20-F FY2025 and 6-K Q3 2025; AerCap Materials factsheets 2023 and 1Q 2026; GE Aerospace, 2026, via D2; FTAI 10-K FY2025; PRNewswire, 2025 (title); GA Telesis releases, 2025; ePlaneAI; AviTrader; Asian Aviation; GlobeNewswire, 1 Dec 2023. Compiled from D3 §3 and D11 §3. "GTF" is Pratt & Whitney's geared turbofan.

Three structural observations come from the sources themselves rather than from this primer. AerSale describes more companies entering the teardown sector and a hypercompetitive feedstock market. Lessors have internalised the teardown step, AerCap through AerCap Materials from 2021 and Willis through Willis Aero, so their retired assets no longer reach third-party traders; FTAI's exclusive partnership is a third version of the same move. And GA Telesis's "OEM-aligned" purchase shows the manufacturer's sanctioned channel running beside the independents' (§33.6). [AerSale, as relayed; AerCap Materials factsheets; Willis Lease 10-K FY2025; FTAI 10-K FY2025; GA Telesis, Oct 2025; D3 §3]

35.2 HOW BIG THE MARKET IS: THREE VENDORS AND A CAVEAT

EXHIBIT 5.7 — USED-SERVICEABLE-MATERIAL MARKET SIZE, 2025, AS EACH VENDOR STATES IT

Source and level	2025	Later figure	Stated growth rate	Scope as retrieved
The Business Research Company, via a research-aggregator listing (market-research vendor)	\$5.87B	\$6.22B (2026)	6.1%	"Air Transport USM Global Market Report"; scope definition not retrieved

Source and level	2025	Later figure	Stated growth rate	Scope as retrieved
Market.us (market-research vendor)	\$7.7B	\$12.3B (2035)	4.8% (2026–2035)	"Air Transport USM Market"; scope not retrieved
Fortune Business Insights, as relayed in search results (market-research vendor)	\$8.55B	\$8.95B (2026); \$12.93B (2034)	4.7% (2026–2034)	Report page not isolated; scope not retrieved
Safe Fly Aviation (consultancy blog)	">\$4B/yr, CFM56 largest segment"	None	None	Scope not stated

Source: The Business Research Company via a research-aggregator listing (GII), 2026; Market.us, "Air Transport USM Market", 2025–26; Fortune Business Insights, figures as relayed, URL not isolated; Safe Fly Aviation, 2026, via D2 §4.6. Reconciliation R-021; D3 tension 1 and unknown U-D3-18.

The three vendor figures for the same year differ by 46% from lowest to highest, and their growth rates differ too, 6.1% against 4.7–4.8%. None of the scope definitions was retrieved: what counts as used serviceable material, whether engines, airframes and components are all included, and whether the figures are sales or gross proceeds. The primer carries all three with the consultancy figure labelled as such and does not average or choose; the scope question is unknown U-D3-18. [reconciliation R-021; D3 tension 1, unknown U-D3-18] For a sense of proportion only, and not as a reconciliation, AerSale's used-material revenue of \$137.6M is between 1.6% and 2.3% of the three market figures, and AAR's derived Parts Supply segment of about \$1,489M, most of which is distribution of new parts, is larger than any one of the three estimates is of used material alone.

36. The module as a unit of trade

Part I §3 boxed the module: a section of the engine designed to be removed and replaced as a unit, bolted at flanges, with its own part number, serial number and maintenance record. Part I §8 boxed the idea this section carries in full, the module as a unit of trade, and its companion, the sum of modules. This section explains how a serviceable module from a teardown is valued, how it is sold, and how it feeds the exchange business that Part VII §46 owns.

36.1 WHICH LEVEL IS BEING COUNTED

The CFM56-7B's major modules are the fan and booster, the core (the high-pressure compressor, combustor and high-pressure turbine), the low-pressure turbine (LPT), and the accessory gearbox, each built from numbered sub-modules whose count the research did not reach. That is the anatomical level Part I §3 fixed. FTAI's own releases count three modules per engine, the three gas-path modules without the gearbox: "450 modules (150 engines)" and "1,800 modules (600 engines)". The two counts describe the same hardware at different levels, and this Part names the level each time it converts: a torn-down engine yields four major modules anatomically and three in FTAI's counting convention. [Part I §3.2, Exhibit 1.4; FTAI release, 26 Feb 2025, via orchestrator notes; reconciliation R-035, B-3; E.1 rule 7]

36.2 HOW A MODULE FROM A TEARDOWN IS VALUED

A serviceable module from a torn-down engine is valued the same way a whole engine is, in miniature. Its value is the sum of its serviceable parts (airfoils, cases, bearings, seals) at the used-material discounts to list of §34, plus the remaining life of the life-limited parts inside it (life fraction times list price, discounted), less the cost of any repair needed to make it serviceable as a unit. A core module with turbine and compressor disks at 80% life is worth far more than one at 20% life, because the next owner's shop visit will or will not have to open it to replace them. And its documentation must show the history of every life-limited part inside it, so §31 applies to the module as a whole. [D3 §2.7]

EXHIBIT 5.8 — VALUING A SERVICEABLE MODULE FROM A TEARDOWN: THE COMPONENTS, THE PRICING BASIS AND WHAT THE RESEARCH COULD NOT SOURCE

Component of value	Pricing basis	Figures reached	Source and level	Status
Life-limited parts inside the module	Σ (cycles remaining $\div$ limit) $\times$ list price, then the used discount	One -7B part: HPT disk \$420,000 new, 20,000-cycle limit; full -7B set \$5.7M (Jun 2025) or about \$4M (Nov 2019); -5B set \$9.88M full life, basis unstated	Safe Fly Aviation, 2026 (consultancy blog); MyAirTrade (market-data site); Aircraft Value News (trade press); Calver, Apr 2023 (conference)	Per-part list prices unknown (U-D3-06); core group's share of the set unknown (U-D1-08)
Serviceable non-life-limited parts	Used discount to list: 60–80%, or 70–75% for repaired engine material sold to an airline	Figures in §34	Aviation Week; Aircraft Commerce Issue 120 (trade press)	Sourced, two readings (R-022)

Component of value	Pricing basis	Figures reached	Source and level	Status
Repair needed to make the module serviceable as a unit	Repair-shop quote	Not captured	None	Unknown (U-D3-09 covers teardown cost; module repair cost not separately logged)
Whether a module change requires a test-cell run under the shop manual	Shop manual procedure	Not verified	None	Unknown (U-D1-14)
Premium for avoided downtime when sold through an exchange	Exchange fee plus the difference in life-limited-part life	No provider's exchange price list was reached, including FTAI's	None	Unknown (K-01: U-D2-07, U-D3-20, U-D5-03)
Whole-module reference	A serviceable module sold to an engine shop or exchange pool "almost immediately" after teardown	No module price was captured	D3 §2.2	Timing sourced; price unknown

Source: D3 §2.7 and §2.8 (the valuation structure, assembled in the research); Safe Fly Aviation, 2026, via D2 §4.4; MyAirTrade, Jun 2025; Aircraft Value News, Nov 2019; Mark Calver, Cargo Facts, Apr 2023; Aviation Week and Aircraft Commerce as in Exhibit 5.5; reconciliation K-01, R-016, R-022; D1 unknowns U-D1-08 and U-D1-14. "HPT" is the high-pressure turbine.

The exhibit shows why §36 cannot price a module to the dollar. The two unknowns that would let it, the list price of each life-limited part in the core and any provider's exchange fee, were not reached. What can be shown is the structure, and the one engine-level cross-check the research supports.

WORKED EXAMPLE 5.6 — THE LIFE-LIMITED-PART LEG OF A CORE MODULE, AND THE ENGINE-LEVEL CROSS-CHECK

Inputs: a -7B core module's life-limited parts are the HPC stage 1–2 spool, the HPC stage 3 disk, the HPC stage 4–9 spool and the HPT disk, all with a 20,000-cycle limit (sourced: records mini-packs, Part I Exhibit 1.5). The HPT disk lists at \$420,000 new (sourced: Safe Fly Aviation, 2026,

consultancy blog). The other three list prices are unknown (U-D3-06), and the core group's share of a full set is unknown (U-D1-08). Average life remaining in the core 40% (assumed). Used discount 70–75% of list (sourced: Aircraft Commerce Issue 120).

HPT disk alone: $0.40 \times 420,000 = \$168,000$ life-proportional; $\times 0.70$ to $0.75 = \$117,600\text{--}126,000$
 the other three core parts: $0.40 \times (\text{unknown list}) \times 0.70$ to $0.75 =$
 cannot be stated
 core module life-limited-part leg = $117,600\text{--}126,000 + (\text{three unpriced parts}) =$ a floor, not a value

Because the core's share of the set is unknown, the example steps up to the engine. An engine whose life-limited parts average 40% remaining (assumed) carries, on each of the three set prices never mixed in one line:

on the -7B set of \$5.7M (sourced: MyAirTrade, Jun 2025, scope not visible): $0.40 \times 5.7\text{M} = \$2.28\text{M} \rightarrow$ at 70–75%: $\$1.60\text{--}1.71\text{M}$
 on the -7B set of \$4.0M (sourced: Aircraft Value News, Nov 2019): $0.40 \times 4.0\text{M} = \$1.60\text{M} \rightarrow \$1.12\text{--}1.20\text{M}$
 on the -5B figure read as per engine, \$4.94M (sourced: Calver, Apr 2023; basis unknown, U-D3-03): $0.40 \times 4.94\text{M} = \$1.98\text{M} \rightarrow \$1.38\text{--}1.48\text{M}$

Set against the consultancy's \$1.6–2.2M used-material yield for a run-out engine, the life-limited-part leg at 40% is between half and all of it, before any airfoil, case or accessory is counted. That is why the trade advertises used life-limited parts on cycles remaining and why the module that holds the shorter-lived parts, the core, is the one whose remaining life moves the most money. The mismatch Part I §8 described is the other side of the same arithmetic: a fan module at 18,000 of 30,000 cycles (60%) attached to a core at 1,500 of 20,000 (7.5%) is worth more as a fan sold to someone who needs a fan plus a core sold for its stub life than as one engine that must go into a shop, because the buyer of the fan pays for 60% of the fan's list and the buyer of the core pays only for 7.5% of the core's. Limits of the example: one list price out of four; the life fraction is assumed; the three set prices disagree (R-016); and the non-life-limited parts, the repair cost and any exchange premium are not priced. [D3 §2.7–2.8; D1 §2.8; Safe Fly Aviation, 2026 (consultancy blog); MyAirTrade, Jun 2025; Aircraft Value News, Nov 2019; Calver, Apr 2023; Aircraft Commerce Issue 120; reconciliation R-016, R-022; D1 unknown U-D1-08]

36.3 HOW A MODULE IS SOLD, AND HOW IT FEEDS AN EXCHANGE BUSINESS

A serviceable module from a teardown has three kinds of buyer. An engine shop buys it to install in a customer's engine in place of a module that has run out, which turns a full disassembly into a module change (Part I §3.4). A trader buys it to hold or to break into parts. And an exchange provider buys it, or produces it from its own feedstock, to put into a pool. The first and third can be placed almost at once after teardown, which is why §32.3 said the engine proceeds arrive fastest. [D3 §2.2, §2.7]

The exchange is the transaction that turns a module into a product rather than a part. An engine shop or operator hands over a time-expired or damaged module and receives a serviceable one, paying an exchange fee plus the difference in life-limited-part life between the two. The incoming module becomes the provider's feedstock: the provider tears it down, salvages the serviceable parts and the life, rebuilds, and returns it to the pool. For the shop it converts a full shop visit (open the whole engine, replace parts, re-assemble, test) into a shorter module swap. For the provider it is a way to sell used material and life-limited-part life at module prices, which carry the premium of avoided downtime. Part I §3 boxed module exchange; Part VII §46 and §47 own FTAI's version and its pricing; GE sells the same thing under TrueChoice Transitions (§33.6). [D3 §2.7; D2 §2.3; Part I §3.4]

FTAI's filings describe this chain directly, in the language §35.1 quoted: engines unserviceable for leasing are torn down "into salvageable components that would be manufactured into new modules and parts and sold", and the business sells "engines, modules and parts" under the name Maintenance, Repair and Exchange (MRE). [FTAI 10-K FY2025; D5 §2.2] What no provider publishes is the exchange price. The fee above the life transferred was not found for any provider, including FTAI, which is the cluster K-01 that three dossiers raised; the only bound the research holds is FTAI's products revenue divided by its modules produced, \$2.11M for 2025, which Part VII §56 presents as an upper bound and not a price because the numerator includes whole engines, parts and the fair value of the units handed back. [reconciliation K-01; E.1 rule 11; Part VII §47, §56]

THE TWO MEANINGS OF "CORE"

The word carries two meanings in this primer and the registry flags the conflict. In the anatomical sense, used throughout Parts I to VI, the core is the gas generator: the high-pressure compressor, combustor and high-pressure turbine, the hot module whose disks have the shortest life limits and which therefore dominates the life-limited-part arithmetic above. In the exchange sense, used by FTAI's segment and in Part VII, the core is the unserviceable unit a customer hands back in an exchange, which has value to the exchanger as feedstock and is measured at fair value as noncash consideration. A sentence such as "the core is worth more at 80% life" is about the first; "the exchange fee plus the core" is about the second. This Part writes "core module" or "gas generator" for the first and "returned core" or "the unit handed back" for the second. [registry C-6; D1 §2.3; D5 §2.3; Part VII §47]

Two constraints limit which torn-down modules fit which pools, and both come from the manual rather than the market. A -7B24 and a -7B27 differ in value, \$5.7M against \$6.4M on the appraiser's half-life market values, and the interchange of modules and life-limited parts across thrust ratings and across the -5B and -7B is governed by the shop manual; the -5B and -7B share enough core hardware that dealers advertise a single "CFM56-5B & -7B Core LLP Package", but

the exact list of common parts was not retrieved. [IBA, Sept 2025; Part I §3.3; D3 §5; D1, unknown U-D1-02] Build standard and records, the conditions Part I §3.3 set for a module to move, apply to a module from a teardown exactly as to one from a shop.

The half-life trade of §32.5 closes here. A part-worn engine is leased for its months or torn down; a module from the teardown is valued on the life inside it; the life-limited parts are priced at their fraction of list less a discount; and the module goes into a pool from which an exchange sells that life back to an operator at a price that also buys time. Each step prices the same thing, remaining life, in a different wrapper. Part VII §46 takes the last step in full.

Tensions carried in this Part

Id	The two (or more) figures	Where in this Part
R-001	CFM56 in service: ~23,000 (CFM, undated); ~24,000 (Aviation Business News, Sept 2025); >22,800 (Safran, end-2025); consultancy ~14,200 and >19,000 (Safe Fly)	§33.3, worked example 5.4
R-002	Safe Fly's ~14,200 read as the -7B fleet (D3) or all CFM56 (D8)	§33.2, §33.3
R-004	CFM56 retirement rate: GE 3–4% → 2–3% → 1.5–2% (2026 planning); ~1.5% 2025 actual, ~2% 2026 expected; <1% in 1Q 2026; Safe Fly "near 2%"; McKinsey 24% below norm, 2.7% from 2028	§33.2, Exhibit 5.4; worked example 5.4
R-005	Aircraft retirements 2024: "fewer than 200" (IBA) against 587 (Cirium); Cirium 502/465/333/325 (2019–22); McKinsey-implied ~410 a year against Cirium-projection-implied ~1,130 a year	§33.2, Exhibit 5.4
R-008	Interval: "roughly every eight years", 24-year de facto retirement (Aviation Week) against 10,000–15,000 cycles at 1,000–1,500 a year (Leeham; Aircraft Value News)	§33.1
R-016	Full -7B LLP set: \$5.7M (MyAirTrade, Jun 2025) against ~\$4M (Aircraft Value News, Nov 2019); 2013 \$1.7M (caveat); Calver -5B \$9.88M full life, basis unstated	§32.6 worked example 5.3; §36.2 worked example 5.6; Exhibits 5.3, 5.8
R-018	Escalation: spares ~10%/yr, double-digit Nov 2022, high-single-digit Aug 2023, inflation +3–4 points (Aviation Week; Safran) against LLP set ~7%/yr derived, Ishka ~12% step, -7B parts +20–30% in 2023 (Magnetic)	§34.3, worked example 5.5

Id	The two (or more) figures	Where in this Part
R-020	Parts constraint: GE 10-K material input +40%, turnaround +10% against Oliver Wyman labour shortage and AVM "USM supply remains thin"	§33.4
R-021	USM market 2025: \$5.87B (The Business Research Company); \$7.7B (Market.us); \$8.55B (Fortune Business Insights); ">\$4B" (Safe Fly)	§35.2, Exhibit 5.7
R-022	USM price: 60–80% of new, up to 85%, above list (Aviation Week) against 70–75% of list, trader to airline (Aircraft Commerce Issue 120)	§34.1, Exhibit 5.5; worked example 5.5
R-028	-7B rent: ~\$100,000/month (IBA, Apr 2024) against \$42,000–48,000 (Safe Fly, 2026, may be green-time)	§32.5 worked example 5.2; §32.6; §33.4
R-029	-7B value: half-life market value \$5.7M/\$6.4M, "levelled off" (IBA) against green-time \$2.8–3.4M "(new condition)", teardown \$0.8–1.2M, rents +15–20% (Safe Fly) and values +50% in two years (ePlaneAI)	§32.5 and its callout; Exhibit 5.3
R-030	Lease rate factor 0.70%/month derived on mixed bases (D3); no stated figure	§32.5 worked example 5.2
R-033	AerCap engines: 470 owned (20-F) against >1,200 including SES (6-K)	§35.1
R-034	Freighter conversion: >260 737-800Fs in service, pipeline >300 (Cargo Facts; Acumen) against IBA's "oversupply of narrowbody conversions" (Sep 2026)	§32.6, closing paragraph
R-035	Module count: four major modules (Part I anatomy) against three per engine (FTAI releases)	§36.1
D3 tension 1	The three vendor market sizes differ by 46%; scopes not retrieved	§35.2
D3 tension 2 (with R-004, R-005, B-6)	Oliver Wyman's 2020 "tsunami of used aircraft parts" (title only) against the 2022–2026 scarcity described by Cirium, McKinsey, GE, AerSale, IBA and Aviation Week	§33.5
D3 tension 5	Same content as R-022	§34.1
D3 tension 6	AerSale's "hypercompetitive feedstock" (buyer's view), IBA's "lessors' market" and Aviation Week's "seller's market" (owner's view): same direction, different vantage	§33.4
D3 tension 7	Direction of USM prices 2024–26: Aviation Week "stabilizing" and "upturn" (undated); IBA "levelled off" (H1 2026); Safe Fly rents +15–20% "in recent months"	§33.5

Unknowns carried in this Part

Entries from the USM dossier (D3), verbatim from the reconciliation register, followed by the entries from other dossiers that the register clusters with them (K-01, K-02, K-15, K-16). Where the orchestrator notes partly resolved an entry, the resolving fact is attached after it.

- **U-D3-01 (D3) — U1** — Sourced split of a 737-800 part-out value between engines and airframe. Would be resolved by an appraiser part-out report or an AerSale/AAR investor presentation with a teardown waterfall.
- **U-D3-02 (D3) — U2** — Total sell-through period for teardown inventory (months to recover, say, 80% of proceeds). AerSale's 10-K inventory-turnover disclosure or a trade-press teardown case study would answer it.
- **U-D3-03 (D3) — U3** — Whether Calver's \$9,878,733 / \$4,939,366 CFM56-5B LLP figure is per engine or per aircraft (two engines). The original Cargo Facts presentation would settle it.
- **U-D3-04 (D3) — U4** — Any mba Aviation (or Ascend by Cirium) CFM56-5B/-7B value or lease-rate figure. None returned.
- **U-D3-05 (D3) — U5** — A current appraiser half-life value for the CFM56-5B (only trend commentary and a trade-press lease-rate range were captured).
- **U-D3-06 (D3) — U6** — Current CLP of representative CFM56-7B LLPs (e.g. HPT disk, fan disk) and of a full -7B LLP set. Needed for a precise module-value example.
- **U-D3-07 (D3) — U7** — The identity and terms of FTAI's "exclusive partnership" for teardown, repair, marketing and sale of CFM56 parts. The 10-K excerpt does not name it. The FTAI 10-K text or an FTAI press release would answer it. *Partly closed (reconciliation C.3 item 1, B-9): a 2025 PRNewswire release is titled "AAR and FTAI Aviation extend their exclusive Serviceable Engine Products agreement, providing CFM56 engine material to the global aviation aftermarket through 2030"; title located, body not read; terms still open.*
- **U-D3-08 (D3) — U8** — Purchase price of a 737-800 bought for part-out in 2025–2026 (and of the airframe alone).
- **U-D3-09 (D3) — U9** — Teardown labour and facility cost per aircraft and per engine.
- **U-D3-10 (D3) — U10** — Values of a 737-800 APU, avionics suite, thrust reversers and other rotables.
- **U-D3-11 (D3) — U11** — AAR FY2026 total revenue and Parts Supply segment revenue in dollars (only the share and growth were captured). *Partly closed (reconciliation C.3 item 6): D11 derives consolidated sales of \$3,308.0M and Parts Supply of about \$1,489M from the FY2026 10-K; the filing's own segment figure was not returned.*

- **U-D3-12** (D3) — **U12** — Willis Lease portfolio value, number of engines, and Willis Aero spare-parts revenue for 2025. *Partly closed (reconciliation C.3 item 5): 363 engines, 20 aircraft, \$2,801.7M of equipment held for lease, 84.9% utilisation (Willis Lease 10-K FY2025, via D11); Willis Aero revenue still open (K-15).*
- **U-D3-13** (D3) — **U13** — AJW Group scale.
- **U-D3-14** (D3) — **U14** — Unical Aviation: nothing returned. *Partly closed (reconciliation D.3): Unical Aviation was FTAI's JV partner in QuickTurn Miami (GlobeNewswire 1 Dec 2023; 10-K). Unical's own USM scale: still open.*
- **U-D3-15** (D3) — **U15** — GE Aerospace's used-material arm: name, scale and CFM56 role not captured (the only hit was a TrueEngine presentation URL whose content was not read).
- **U-D3-16** (D3) — **U16** — Actual retirement counts for 2023, 2024 and 2025 from Cirium (only 2019–2022 counts and the 2024–26 shortfall percentage were captured). *Partly closed (reconciliation C.3 item 4): IBA "fewer than 200" (2024) and Cirium 587 (2024) via D9, unreconciled (R-005); GE ~1.5% of the CFM56 base in 2025.*
- **U-D3-17** (D3) — **U17** — Number of CFM56-5B and -7B engines torn down per year, 2023–2025.
- **U-D3-18** (D3) — **U18** — Scope definitions behind the three USM market-size estimates (what counts as USM; whether engines, airframes and components are all included).
- **U-D3-19** (D3) — **U19** — Oliver Wyman's 2020 forecast figures and any updated Oliver Wyman USM forecast (title only captured).
- **U-D3-20** (D3) — **U20** — Pricing of a module exchange (exchange fee, LLP-life true-up) from any provider, including FTAI. D5 may hold FTAI's own disclosures.
- **U-D2-07** (D2, cluster K-01) — **U7. Module-exchange pricing.** No source gives the exchange fee or a worked price for a CFM56 HPT or core exchange; Safe Fly's "\$150,000–\$400,000 removal cost per module" is ambiguous. D5 may find it in FTAI's investor materials or the In Practise interview ("FTAI Aviation: CFM56 repair and module swap", URL returned, not read).
- **U-D5-03** (D5, cluster K-01) — **U3 — Exchange turnaround time and exchange pricing.** FTAI's quoted turnaround and any published exchange-fee ranges. Trade press (e.g., the In Practise interview [S38]) likely has estimates; not read.
- **U-D1-09** (D1, cluster K-02) — **U9** Current half-life base values for -7B and -5B by rating (IBA, mba, Cirium). Resolve with an appraiser publication; this session's searches returned definitions, not values.
- **U-D2-08** (D2, cluster K-02) — **U8. Value of a freshly overhauled full-stack CFM56-7B** versus a green-time unit. Needed to close the worked example. Resolve via IBA/Ascend/mba engine value tables.

- **U-D6-09** (D6, cluster K-02) — **U9** — **Half-life market values** for 12–15-year-old 737-800 and A320ceo aircraft, and full-life CFM56-7B/-5B/V2500 values, 2023–2026. Resolve: IBA/Cirium/mba published updates (D3 may hold engine values).
- **U-D11-03** (D11, cluster K-15) — **U3** — **Willis Lease maintenance services and spare-parts revenue lines for FY2025**. The 10-K's revenue table was not returned beyond lease rent, the combined core line and gain on sale. Resolve: read the consolidated statement of income in [S1].
- **U-D4-10** (D4, cluster K-16) — **U10** — **Any CFM/GE statement specifically on the Chromalloy HPT blade PMA**, and any GE "material" programme (e.g. TrueChoice Material) positioning against PMA. The brief's phrase "material solutions" was not matched to a current CFM/GE programme in this session; the historical P&W "Global Material Solutions" CFM56-3 PMA venture was found only via headlines [51][52].

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